

MS&E 246 Final Report

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March 23, 2026

Contents

1	Introduction	1
2	Data Cleaning and Preprocessing	1
2.1	Overview	1
2.2	Removal of Invalid Records	2
2.3	Business Logic and Temporal Consistency	2
2.4	Missing Values and Categorical Processing	2
2.5	Outlier Treatment	3
3	Macro Data Augmentation	3
3.1	Overview	3
3.2	Merging Strategy and Temporal Alignment	4
3.3	Macro Variables and Financial Market Variables	4
3.4	Construction of Dataset	5
4	Exploratory Data Analysis	6
4.1	Overview	6
4.2	Default Rate over Time	6
4.3	Loan Size and the Timing of Risk	7
4.4	Industry Heterogeneity	8
4.5	Geographic Differences	9
4.6	Firm Characteristics: Business Age and Business Type	11
4.7	Macro Indicators and Default Rate	12
4.8	Correlation Structure	12
4.9	Implications for Modeling	13
5	Feature Engineering	14
5.1	Overview	14
5.2	Shared Feature Engineering Pipeline	14
5.3	Train-Validation-Test Split	15
5.4	Model-Specific Target Construction	15
6	Binary Response Models	16
6.1	Overview	16
6.2	Baseline Models	16
6.3	Improved Models with Feature Selection and Hyperparameter Tuning	17
6.4	Distribution Shift Adjustment	19
6.5	Interpretable Logistic Regression	22
7	Survival Models	22
7.1	Overview	22

7.2	Kaplan-Meier Exploration	23
7.3	Static Cox Models	24
7.3.1	Time to Default Model	24
7.3.2	Time to Pay in Full Model	26
7.4	Time-Varying Cox Models	27
7.4.1	Unified Fitting Pipeline	27
7.4.2	Time to Default Model	28
7.4.3	Time to Pay in Full Model	30
8	Portfolio Loss Simulation and Tranche Analysis	32
8.1	Overview	32
8.2	Simulation Pipeline	32
8.3	Risk measures and Tranche Mapping	34
8.4	Numerical Results	34
8.4.1	Equal-EAD Pipeline	35
8.4.2	Dollar-EAD Pipeline	36
8.4.3	Direct Comparison of the Two EAD Specifications	37
9	Conclusion	38

1 Introduction

Small business lending plays a critical role in economic development, yet it is inherently exposed to significant credit risk due to the heterogeneity of borrowers, industries, and macroeconomic environments. In this project, we study credit risk in the context of the U.S. Small Business Administration (SBA) 504 loan program, using a comprehensive dataset of equipment loans spanning from 1990 to 2024. Our primary objective is to develop a rigorous and interpretable framework for modeling loan default behavior and quantifying portfolio risk.

Our modeling strategy integrates data processing, predictive modeling, and portfolio-level risk evaluation into a unified framework.

We begin by constructing a cleaned and feature-engineered dataset tailored for both static and time-varying analyses. We augment the dataset with time-varying macroeconomic and financial variables, including state-level indicators and national market factors. These features allow the model to reflect evolving economic conditions throughout the life of each loan.

Building on this foundation, we first explore a range of machine learning models to predict default probabilities. Particular attention is paid to distributional shifts over time, including both covariate shift and label shift, which are critical challenges in out-of-sample prediction for long historical datasets.

While binary classification provides useful benchmarks, it fails to capture the timing of credit events. To address this limitation, we develop survival models to estimate the hazard of default and prepayment. By incorporating time-varying covariates, these models are able to reflect the dynamic evolution of credit risk, and empirically demonstrate substantial performance improvements over static approaches, especially at medium and long horizons.

Finally, we translate the estimated models into a portfolio risk framework. Using Monte Carlo simulation, we generate the distribution of losses for a hypothetical portfolio of 1,000 loans over 5- and 10-year horizons. Based on the simulated loss distribution, we compute standard risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES), and further evaluate tranche-level risk for mezzanine and senior investors. This allows us to assess how credit losses propagate across different layers of the capital structure and provides a comprehensive view of portfolio credit risk.

Overall, this project provides a comprehensive framework that integrates data cleaning, feature engineering, predictive modeling, and portfolio simulation. It highlights the importance of incorporating time-varying information and realistic default dynamics in credit risk modeling, and offers practical insights for risk management in small business lending portfolios.

2 Data Cleaning and Preprocessing

2.1 Overview

The raw dataset consists of two publicly available SBA 504 loan files covering different time periods: fiscal years 1991–2009 and 2010–present. These two datasets are first vertically concatenated into a unified dataset, resulting in an initial sample of 224,331 loan records.

The primary objective of the data cleaning process is to ensure data quality, consistency, and usability for downstream tasks, including exploratory data analysis (EDA), feature engineering, and predictive modeling. The full pipeline consists of the following steps:

- Removing invalid or inconsistent observations.
- Enforcing business and temporal constraints.
- Handling missing values and encoding categorical variables.
- Treating outliers and exporting the final dataset.

The final cleaned dataset is saved in `parquet` format for efficient downstream processing.

2.2 Removal of Invalid Records

We begin by removing observations that are not suitable for analysis or contain obvious data issues.

Loan status filtering. Loans with status `CANCLD` (canceled) and `NOT FUNDED` are removed, as they do not represent active credit exposures. Observations with missing `loanstatus` are also excluded.

Duplicate removal. Duplicate records are removed, keeping the last observation to ensure that each loan appears only once.

Invalid numeric values. We enforce non-negativity constraints on key financial variables:

- `thirdpartydollars` must be non-negative if present.
- `grosschargeoffamount` must be non-negative.

Minimum loan size. Loans with `grossapproval` $\leq 1,000$ or missing values are removed, as these are likely to be data errors or economically insignificant.

Minimum loan term. Loans with `terminmonths` < 120 months (10 years) or missing values are removed to focus on standard SBA 504 loan structures.

This step removes observations that are either irrelevant for credit risk analysis or clearly inconsistent with the program’s design.

2.3 Business Logic and Temporal Consistency

To ensure internal consistency, we enforce strict business and temporal rules.

Date parsing. All columns containing the string “date” are converted to `datetime` format.

Temporal ordering constraints. We enforce the following ordering relationships:

- `firstdisbursementdate` $>$ `approvaldate`.
- `chargeoffdate` $>$ `firstdisbursementdate` (if present).
- `paidinfulldate` $>$ `firstdisbursementdate` (if present).

Consistency between status and dates. We impose consistency conditions between loan status and event dates:

- PIF loans must have a valid `paidinfulldate` and no `chargeoffdate`
- CHGOFF loans must have a valid `chargeoffdate` and no `paidinfulldate`
- EXEMPT loans should not have both payoff and charge-off dates

These rules are critical for avoiding label inconsistencies and ensuring correct construction of survival times in later modeling.

2.4 Missing Values and Categorical Processing

We next address missing values and high-cardinality categorical variables.

Dropping uninformative columns. The column `deliverymethod` is removed as it contains 100% missing values.

Franchise encoding. Franchise-related variables (`franchisename`, `franchisecode`) are replaced by a binary indicator:

$$\text{iffranchise} = 1(\text{franchise information present})$$

The original columns are then dropped.

Business age recategorization. The `businessage` variable is consolidated into a smaller set of categories: Startup, Young, Established, Ownership, and Unknown.

Third-party lender variables. For `thirdpartylender_name` and `thirdpartylender_city`, only the top 30 most frequent categories are retained, with all others mapped to `Other`. Missing values are handled as follows:

- If `thirdpartydollars` is present, missing lender fields are imputed using the mode.
- Otherwise, missing values are set to `NoThirdParty`.

We also create:

- `ifthirdparty`: indicator for presence of third-party funding.
- Missing `thirdpartydollars` are filled with 0.

Industry classification. The `naicscode` variable is converted to its 2-digit prefix to represent industry sectors. Rare categories (frequency ≤ 100) are grouped into `Other`, and missing values are set to `Unknown`. The corresponding description column is removed.

Other categorical processing.

- `borrcity`: retain top 30 cities, others mapped to `Other`.
- `collateralind`, `businesstype`, `cdc_state`: missing values set to `Unknown`.
- `processingmethod` mapped to standardized short codes.
- `subprogram` consolidated into a smaller number of categories.

Column removal. Columns with limited predictive value or redundancy are removed, including: `cdc_name`, `cdc_street`, `cdc_city`, `cdc_zip`, `projectcounty`, `congressionaldistrict`, `borrstreet`, `program`, `borrzip`, `sbadistrictoffice`, and `approvalfiscalyear`.

2.5 Outlier Treatment

We apply simple but robust rules to mitigate the impact of extreme values.

- `terminmonths` is capped at 300 months (25 years), consistent with the maximum SBA loan term.
- `jobssupported` is capped at 50 (approximately the 95th percentile).

These transformations reduce the influence of outliers while preserving the overall distribution of the data.

After all cleaning steps, the dataset is exported as `data_cleaned.parquet`. The final dataset contains 183,914 observations, compared to the initial 224,331 records, corresponding to a removal of approximately 18.0% of the data.

3 Macro Data Augmentation

3.1 Overview

In addition to loan-level characteristics, we incorporate macroeconomic and financial market variables to capture the external environment in which loans are originated and evolve over time. These variables are essential for modeling credit risk, as default behavior is strongly influenced by both local economic conditions and aggregate market dynamics.

Starting from the cleaned loan-level dataset (`data_cleaned.parquet`), we construct two augmented datasets:

- `data_merged.parquet`: a static dataset with one observation per loan.
- `data_merged_tv.parquet`: a time-varying panel dataset with one observation per loan-month.

The augmentation process integrates:

- State-level macroeconomic variables (e.g., GDP, unemployment)

- National monthly indicators (e.g., CPI, PPI, HPI)
- Financial market variables (e.g., equity returns, interest rates, volatility)

All variables are carefully aligned to ensure that only information available at the decision time (or at each panel time point) is used, thereby avoiding look-ahead bias.

3.2 Merging Strategy and Temporal Alignment

A key challenge in macro data integration is preventing information leakage. To address this, we implement a systematic lagging and alignment strategy.

Yearly variables. State-level yearly variables (e.g., GDP) are merged using a one-year lag:

$$\text{lookup year} = \text{approval year} - 1$$

This ensures that only information available prior to loan approval is used.

Monthly variables. For monthly datasets, we dynamically determine the lag based on data availability. Specifically, for each macro dataset, we compute:

$$\text{lag} = (\text{current year-month}) - (\text{last available year-month})$$

and enforce a minimum lag of one month.

Merge implementation. All monthly merges are performed by:

- Applying forward-filling within each state for state-level variables.
- Shifting the loan date backward by the computed lag.
- Matching on year and month (and state, if applicable).

This design ensures that macro variables are aligned with the information available at each observation time.

3.3 Macro Variables and Financial Market Variables

Macro variables and financial market variables (obtained via the `yfinance` API) are merged to the dataset. Both levels and monthly growth rates are included, where the growth rates are computed as:

$$\text{growth}_t = \frac{x_t - x_{t-1}}{x_{t-1}}.$$

The following types of variables are included:

State-level yearly variables.

- GDP (Gross domestic product): lagged by 1 year.

State-level monthly variables.

- Crime rate: lagged by 2 months.
- Employment/Hrs/Earnings index: lagged by 3 months.
- Unemployment rate: lagged by 3 months.

National monthly variables.

- CPI (consumer price index): lagged by 2 months.
- PPI (producer price index): lagged by 2 months.
- HPI (housing price index): lagged by 3 months.

Financial market variables. Daily data are converted to monthly frequency using end-of-month values. For equity indices, monthly returns are computed using percentage change.

- S&P 500 index: lagged by 1 month.
- Russell 2000 index: lagged by 1 month.
- VIX volatility index: lagged by 1 month.
- 10-year Treasury yield: lagged by 1 month.
- Short-term Treasury yield: lagged by 1 month.

Together, these features provide a comprehensive view of both realized and anticipated economic conditions, supporting the downstream modeling through their impact on borrowers' income, expenses, and financing environment.

3.4 Construction of Dataset

The two augmented datasets differ in their observational structure, as shown in Table 3.1.

Table 3.1: Summary of datasets and their structural characteristics.

Dataset	Unit of Observation	Time Alignment	Number of Rows
Static	One row per loan	Approval date	183,914
Time-varying	One row per loan-month	Monthly panel date	17,653,820

The static dataset captures macro conditions at loan origination, while the time-varying dataset reflects evolving macro conditions throughout the loan lifecycle. The latter is constructed by expanding each loan into a monthly panel to support survival and competing risk models.

End date and event definition. For each loan, we define:

- **end_date:**
 - `chargeoffdate` if default occurs.
 - `paidinfulldate` if fully repaid.
 - `asofdate` otherwise (right-censored).
- **event:**
 - 1 = default (CHGOFF).
 - 2 = paid in full (PIF).
 - 0 = censored (EXEMPT).

Panel expansion. Each loan is expanded into monthly intervals from `firstdisbursementdate` to `end_date`. For each loan-month, we construct:

- `month_start`, `month_stop`.
- `loan_age` (in months).
- `panel_date` (end of month timestamp).

Event timing. The event indicator `event_t` is set to zero for all periods except the final month, where it takes the value of `event`.

Macro merging. All macro variables are merged again at the panel level using `panel_date` instead of `approvaldate`, following the same lagging logic as in the static case.

This results in a fully dynamic dataset where macro variables evolve over time for each loan.

The final datasets are saved as:

- `data_merged.parquet` (static).
- `data_merged.tv.parquet` (time-varying panel).

4 Exploratory Data Analysis

4.1 Overview

This section uses the cleaned and macro-augmented loan-level dataset `data_merged.parquet` to examine broad empirical patterns in SBA 504 loan performance before formal modeling. The goal of the exploratory analysis is descriptive rather than causal: we use the data to identify heterogeneity in default risk across time, loan size, industry, geography, firm characteristics, and macroeconomic conditions. These patterns help motivate the feature engineering choices and model structures developed in later sections.

Here, the default label is defined as

$$\text{default_event} = \mathbf{1}\{\text{loanstatus} = \text{CHGOFF}\},$$

so all default-rate calculations in this section refer to charge-off incidence under that definition.

Table 4.1 summarizes the basic dimensions of the analysis sample. These summary statistics confirm that the data cover a long time period and a broad cross-section of borrowers, which makes both temporal and cross-sectional heterogeneity empirically relevant.

Table 4.1: EDA sample overview.

Metric	Value
Loans	183,914
Approval year span	1991–2025
States	54
NAICS 2-digit groups	22
Overall default rate	6.58%
Median loan amount	411,000

4.2 Default Rate over Time

To study the evolution of default risk over the sample period, we group loans by approval year and compute the mean of `default_event` within each cohort.

Figure 4.1 shows the annual charge-off (default) rate by loan approval year, along with a 3-year moving average to smooth short-term fluctuations.

A clear time trend emerges. Default rates rise from the early 1990s and peak around 2005–2007 at approximately 22%–24%. This peak coincides with the period leading into the Global Financial Crisis (2007–2009), suggesting that loans originated during economic expansions may carry elevated subsequent default risk.

Following the crisis, default rates decline sharply for later approval cohorts. From 2010 onward, default rates fall to below 5% and continue trending downward, reaching near-zero levels for the most recent cohorts. This pattern reflects both improved credit conditions and, importantly, right-censoring: newer loans have had less time to default, leading to downward-biased observed default rates.

The COVID-19 period (2020) does not show a visible spike in default rates. This is likely due to policy interventions and insufficient time for defaults to materialize in recent cohorts.

Overall, this figure highlights strong cohort effects and temporal non-stationarity in default behavior, motivating the use of time-based data splits and the inclusion of macroeconomic variables in the modeling framework.

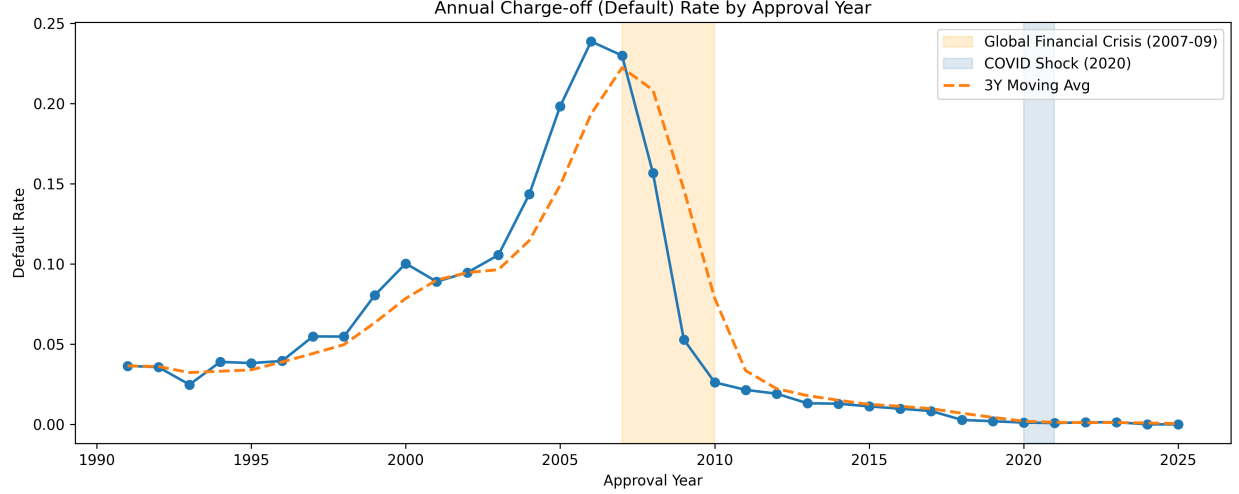


Figure 4.1: Annual charge-off rate by approval year. The solid line shows cohort-level default rates, while the dashed line shows a three-year moving average. Shaded bands highlight the Global Financial Crisis and the COVID shock.

For concreteness, Table 4.2 reports the default rates of the most recent approval cohorts.

Table 4.2: Recent approval cohorts. This table reports observed default rates for recent approval years. The newest cohorts are less seasoned and therefore more heavily right-censored.

Approval Year	Default Rate
2016	0.98%
2017	0.83%
2018	0.28%
2019	0.19%
2020	0.11%
2021	0.08%
2022	0.13%
2023	0.13%
2024	0.00%
2025	0.00%

Therefore, the calendar time matters. This supports the inclusion of macroeconomic and financial variables, and it also supports using time-based train/validation/test splits instead of random splitting.

4.3 Loan Size and the Timing of Risk

Average default rates alone may hide important differences in *when* defaults occur. To examine the timing dimension, we first bin loans by gross approval amount into four groups: <100K, 100K--250K, 250K--500K, and 500K+. We then construct a discrete-time approximation to survival data by computing the number of months from origination to default, payoff, or last observation, and from this calculates (i) a discrete hazard rate and (ii) a cumulative default curve.

Figure 4.2 presents the discrete hazard rates and cumulative default rates across different loan amount bins. The left panel shows that default hazards are relatively low in the early life of the loan and gradually increase over time, peaking around mid-maturity (approximately 40–70 months), before stabilizing or slightly declining. This reflects a typical seasoning effect in credit risk, where defaults are more likely after borrowers have accumulated financial stress over time. In addition, larger loans (500K+) exhibit consistently higher hazard

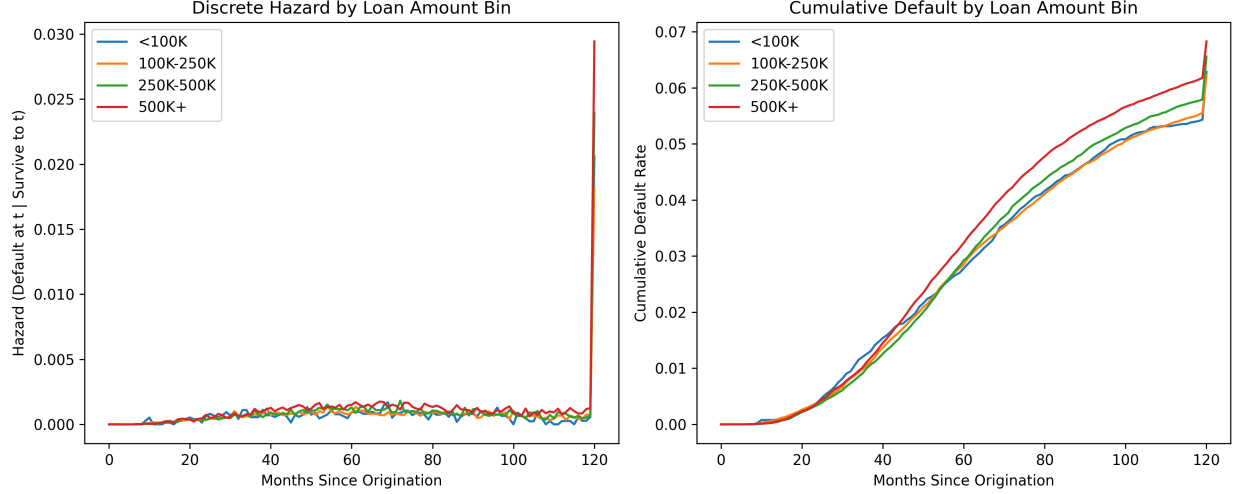


Figure 4.2: Default timing by loan amount bin. The left panel shows the discrete hazard rate, defined as the probability of default at month t conditional on surviving to month t . The right panel shows the cumulative default rate up to month t .

rates compared to smaller loans, particularly during the middle of the loan life. In contrast, smaller loans (<100K.) show lower and more volatile hazard rates. This suggests that larger loans carry higher conditional default risk once they survive the initial periods.

The right panel of Figure 4.2 shows that larger loans (500K+) accumulate the highest default rates, while smaller loans remain slightly lower. Interestingly, the separation between loan size groups is relatively modest in the early periods but widens steadily over time. This indicates that loan size has a stronger impact on long-term risk accumulation than on early default behavior.

There is a sharp spike in both panels at the terminal horizon (around 120 months) is observed across all groups. This is likely a data artifact caused by right-censoring or truncation, where many loans reach the observation boundary, leading to instability in estimation.

This result is consistent with the intuition that larger loans may involve more complex projects, higher leverage, or greater exposure to macroeconomic conditions, making them more vulnerable over longer horizons.

Table 4.3 reports the sample size and average default rate for each loan-size bucket.

Table 4.3: The number of loans and average charge-off rate in each gross-approval bucket.

Loan Amount Bin	Count	Default Rate
< 100K	7,733	6.28%
100K–250K	45,587	6.23%
250K–500K	54,285	6.56%
500K+	76,309	6.83%

Table 4.3 suggests that larger loans have a slightly higher average default rate in this sample. This motivates later modeling choices in two ways. First, loan amount should be included as a predictor. Second, default timing should be modeled explicitly rather than collapsed into a single binary outcome.

4.4 Industry Heterogeneity

We map the raw NAICS codes to two-digit industry sectors and then study default behavior by sector. To improve readability and reduce small-sample noise, the heatmap keeps only the 12 largest sectors by sample size. For each remaining sector and approval year, we compute the mean default rate.

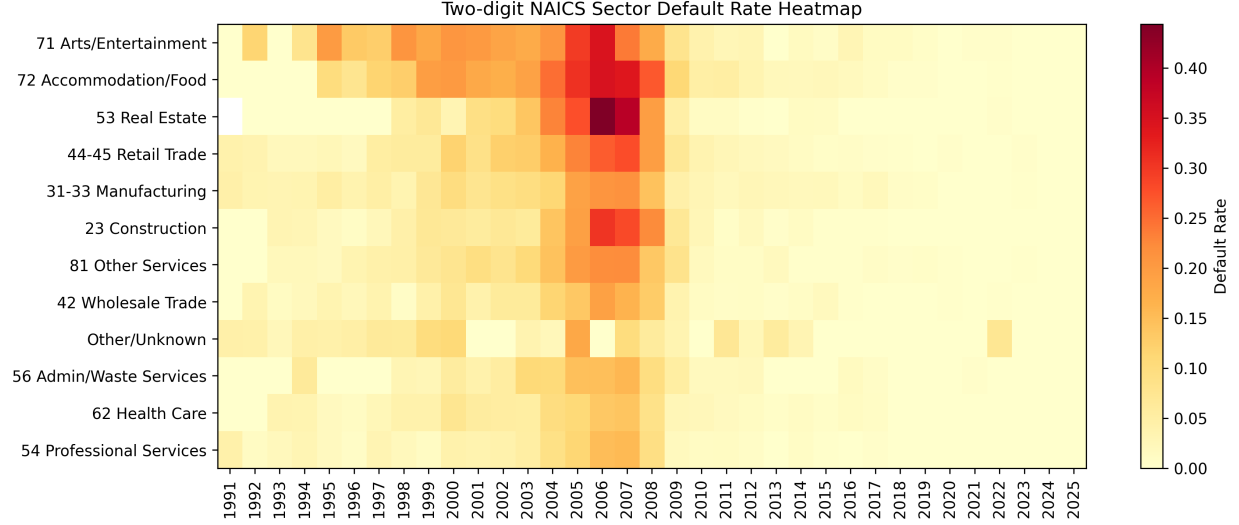


Figure 4.3: Two-digit NAICS sector default-rate heatmap. Rows correspond to large sectors, columns correspond to approval years, and color intensity represents the mean charge-off rate.

Figure 4.3 shows that industry effects are both substantial and persistent. Some sectors appear systematically riskier than others, and the heatmap also reveals that sector risk is not perfectly stable over time. This is more informative than a simple cross-sectional ranking because it shows whether elevated risk is a persistent industry characteristic or tied to specific economic periods.

To complement the heatmap, Table 4.4 reports several of the highest-risk sectors.

Table 4.4: Selected high-risk sectors.

Sector	Count	Default Rate
72 Accommodation/Food	22,694	12.03%
71 Arts/Entertainment	4,499	9.76%
44-45 Retail Trade	20,906	8.15%
52 Finance/Insurance	2,904	7.95%
51 Information	1,356	7.08%

The main insight is that industry is a great determinant of credit risk. In particular, accommodation/food, arts/entertainment and retail stand out as structurally riskier segments in this sample. This directly motivates including industry variables in the later classification and survival models.

4.5 Geographic Differences

We next aggregate default rates by borrower state, producing both a top-versus-bottom bar chart and a choropleth map.

Figure 4.4 shows the ten highest-default and ten lowest-default states in the sample, indicating visible geographic heterogeneity in default behavior. This is consistent with the idea that local economic conditions matter for small-business credit performance.

On the lower end of the spectrum, states such as Massachusetts (MA), Rhode Island (RI), Wyoming (WY), Alaska (AK), and Washington (WA) exhibit relatively low default rates, generally below 4%. These states tend to have more stable economic environments or stronger credit performance in the sample, although the differences are not as pronounced as those observed among the highest-default states.

At the same time, extreme values should be interpreted cautiously when they are based on small sample sizes.

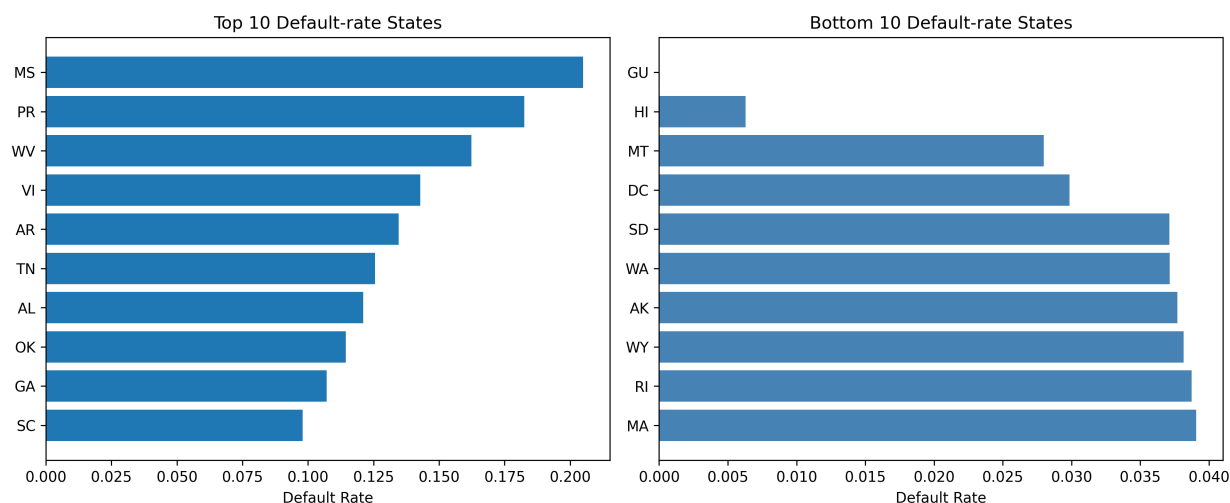


Figure 4.4: Borrower-state default rates. The left panel shows the ten states with the highest average charge-off rates and the right panel shows the ten states with the lowest average charge-off rates.

Figure 4.5 presents the geographic distribution of average default (charge-off) rates across U.S. states, while Table 4.5 highlights selected states with the highest observed default rates.

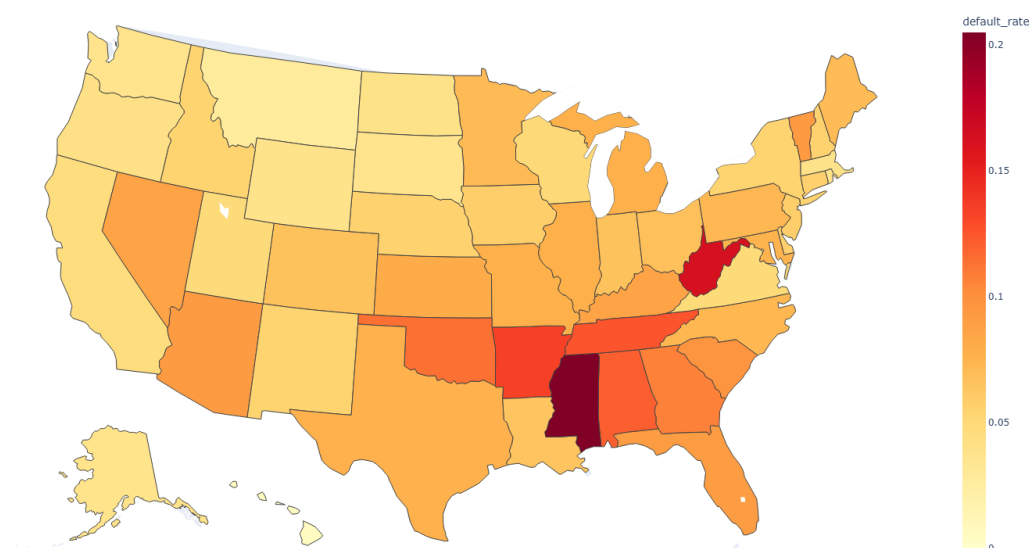


Figure 4.5: Geographic distribution of average loan charge-off (default) rates across U.S. states. Darker colors indicate higher default rates, with elevated risk concentrated in the Southeast and certain economically distressed regions.

A clear regional pattern emerges. Default rates are generally higher in the Southeast and certain economically distressed regions, with Mississippi (MS) exhibiting the highest rate at over 20%, followed by Puerto Rico (PR) and West Virginia (WV). Neighboring states such as Arkansas (AR), Tennessee (TN), and Alabama (AL) also show elevated default levels, suggesting potential regional clustering of credit risk.

In contrast, many states in the Midwest and West exhibit relatively lower default rates, indicating more stable credit performance in those regions.

However, interpretation of extreme values requires caution. Some regions, such as the Virgin Islands (VI) and West Virginia (WV), have relatively small sample sizes, which can lead to high variability in estimated

default rates. As shown in Table 4.5, VI has only 7 observations, making its estimated rate less reliable.

Overall, this figure highlights substantial geographic heterogeneity in default risk, supporting the inclusion of location-based variables and regional macroeconomic indicators in the modeling framework.

Table 4.5: Selected high-default states.

State	Default Rate	N_{obs}
MS	20.49%	410
PR	18.25%	1,441
WV	16.23%	154
VI	14.29%	7
AR	13.46%	609
TN	12.56%	1,067
AL	12.10%	2,396

4.6 Firm Characteristics: Business Age and Business Type

We also examine the heterogeneity across firm age and legal business form. Figure 4.6 shows average default rates for the cleaned business-age categories and for the three main business types.

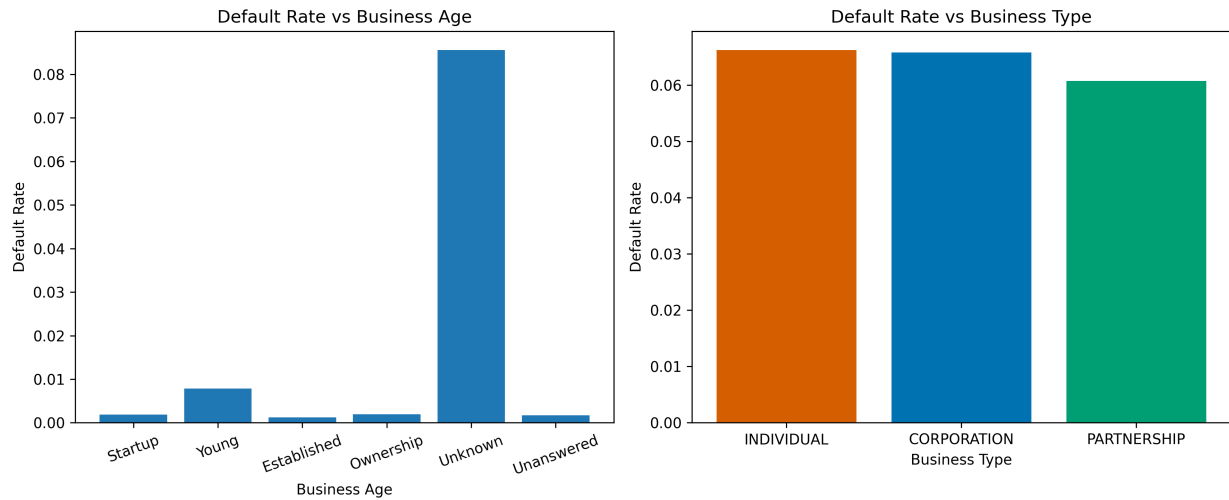


Figure 4.6: Default rates by firm characteristics. The left panel shows average charge-off rates by business age category, and the right panel shows average charge-off rates by business type.

Figure 4.6 suggests that there is heterogeneity across both business age and business type, although the patterns are not equally robust. In particular, the business-age distribution is dominated by the **Unknown** category, which is partly explained by the fact that the SBA only began collecting detailed business age information starting in fiscal year 2018. As a result, some of the apparent differences may reflect reporting limitations or missingness rather than purely economic effects.

Table 4.6 reports the business-age breakdown.

Table 4.7 reports the business-type comparison.

The main takeaway is that firm characteristics do contain useful information, but their reliability differs across variables. Business type appears usable as a standard categorical feature, whereas business age should be modeled more carefully because missingness is pervasive and may itself carry signal.

Table 4.6: Default rates by business age.

Business Age	Default Rate	N
Unknown	8.56%	140,518
Young	0.78%	767
Ownership	0.19%	1,051
Startup	0.19%	4,822
Unanswered	0.17%	576
Established	0.12%	36,180

Table 4.7: Default rates by business type.

Business Type	Default Rate	N
INDIVIDUAL	6.63%	17,819
CORPORATION	6.58%	160,323
PARTNERSHIP	6.08%	5,266

4.7 Macro Indicators and Default Rate

To explore the connection between default risk and the macro environment, we aggregate the loan data to the monthly approval cohort level. For each approval month, we compute the mean default rate and the mean value of selected macro variables already merged into the loan-level data. These series are then standardized using z -scores to make their movements comparable despite different physical units.

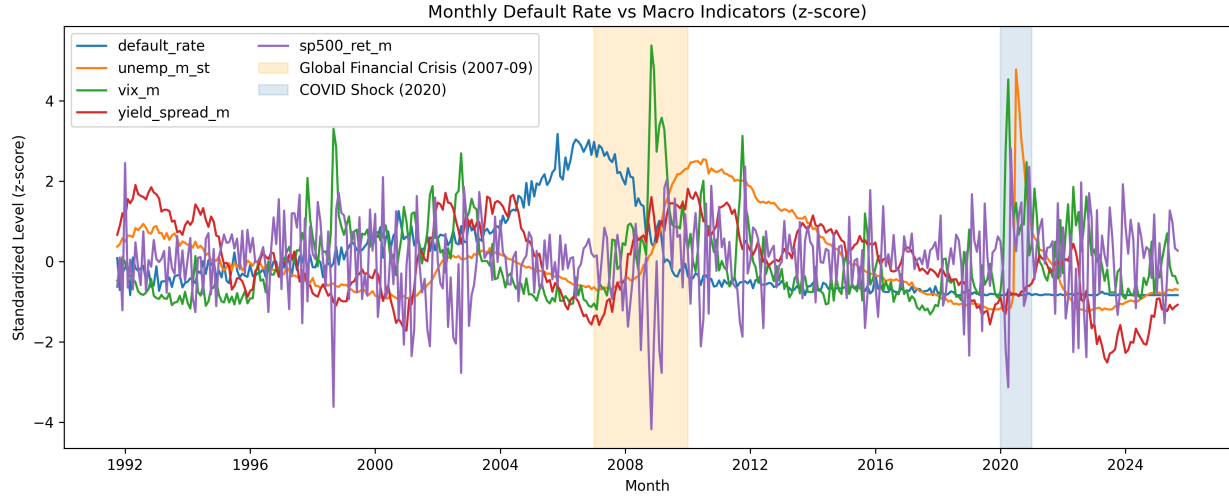


Figure 4.7: Monthly default rate and selected macro indicators. All series are standardized to z -scores, so the plot is intended to show co-movement and direction rather than raw levels. Shaded bands again mark the Global Financial Crisis and the COVID shock.

Figure 4.7 is mainly a visual diagnostic. Because all variables are standardized, the emphasis is on whether variables move together rather than on their original scales. The plot suggests that the monthly default rate co-moves with some macro and financial indicators, especially around major stress periods. This provides preliminary support for augmenting loan-level models with macro variables instead of treating defaults as purely idiosyncratic.

It is worth noting that, due to data availability constraints and to avoid data leakage, macroeconomic variables are merged with a lag (as detailed in Section 3). As a result, the series may not appear perfectly synchronized in the figure. Nevertheless, the overall patterns still indicate co-movement, particularly during periods of economic stress, suggesting that macro conditions play an important role in driving default dynamics.

4.8 Correlation Structure

We also compute the contemporaneous correlation matrix of the monthly default rate and the selected macro variables. Figure 4.8 visualizes that matrix.

Figure 4.8 helps summarize which macro variables are most closely associated with default risk at the monthly

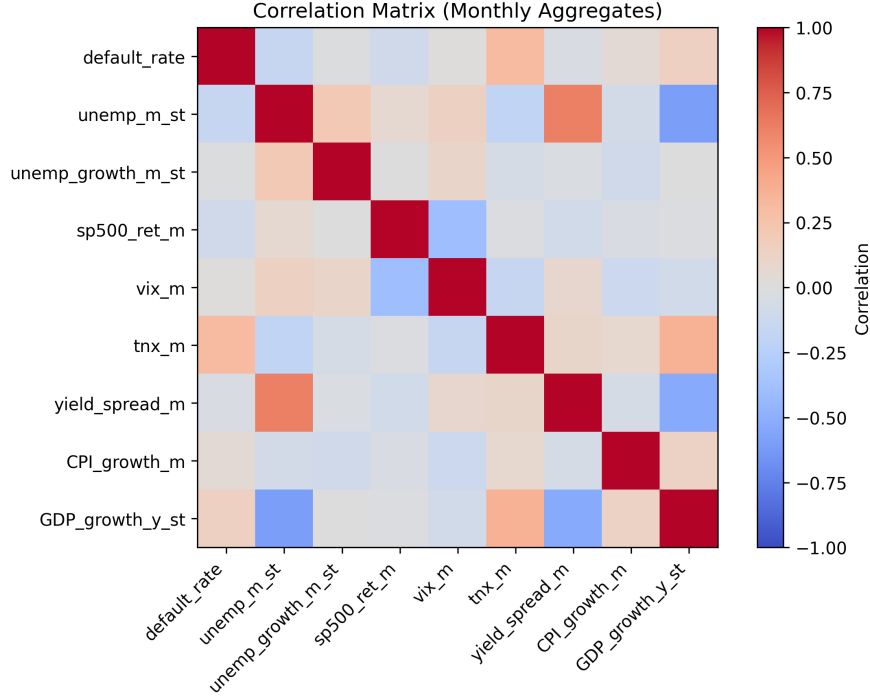


Figure 4.8: Correlation matrix of monthly default rates and selected macro indicators. Correlations are computed using monthly cohort aggregates and therefore should be interpreted as descriptive summary statistics rather than formal structural estimates.

aggregate level. Overall, the correlations among macro variables are moderate rather than excessively high, suggesting that multicollinearity is unlikely to be a severe concern when incorporating these variables jointly into the models. This supports the inclusion of multiple macro indicators to capture different dimensions of economic conditions.

At the same time, the heatmap shows that the pairwise linear correlations between macro variables and the default rate are relatively weak. However, this should not be over-interpreted. The correlation matrix only captures linear relationships, while the true relationship between macro conditions and default risk may be nonlinear or involve interactions. Indeed, as demonstrated in Sections 6 and 7, incorporating these macro variables substantially helps in predictive performance, indicating that they contain useful information beyond simple linear correlation.

4.9 Implications for Modeling

Several empirical patterns emerge from this exploratory analysis.

First, default risk varies meaningfully across approval cohorts, indicating that the external environment matters and that temporal splits are more appropriate than random splits. Second, loan size affects not only average default incidence but also the *timing* of risk realization, which argues for time-to-event methods rather than purely static binary classifiers. Third, industry and geography show large and persistent differences in observed default rates, supporting the inclusion of industry indicators and borrower-state macro variables. Fourth, firm characteristics such as business type and business age contain signal, but some variables must be interpreted with caution because missingness is substantial. Finally, macro indicators exhibit visible co-movement with aggregate default behavior, which motivates the macro augmentation step and the use of time-varying covariates in later survival models.

Overall, the EDA results justify the modeling strategy adopted in the remainder of the report: a combination of cross-sectional prediction, time-varying survival analysis, and macro-aware portfolio risk simulation.

5 Feature Engineering

5.1 Overview

This section describes the feature engineering pipeline used to transform the merged loan-level and macroeconomic data into model-ready datasets. The design aims to support three modeling frameworks: (i) binary classification for default prediction, (ii) static survival models for time-to-event analysis, and (iii) time-varying survival models based on panel data. All steps are implemented to ensure reproducibility and to avoid data leakage across training, validation, and test sets.

The feature engineering process takes as input two primary datasets:

- Static loan-level data: `data_merged.parquet`.
- Time-varying panel data: `data_merged_tv.parquet`.

5.2 Shared Feature Engineering Pipeline

Target construction. The loan status variable is mapped into a binary default indicator:

- CHGOFF $\rightarrow 1$ (default)
- PIF $\rightarrow 0$ (non-default)
- EXEMPT $\rightarrow 0$ (treated as non-default; right-censored in survival models)

For binary classification, only CHGOFF and PIF observations are retained, while survival models additionally include EXEMPT observations as censored cases.

Loan amount and behavioral features. To account for inflation and reduce skewness, loan amounts are first converted into real terms using CPI adjustment:

$$\text{grossapproval}^{\text{real}} = \frac{\text{grossapproval}}{\text{CPI}_m} \times 100.$$

Similarly for third-party contributions. Log transformations are then applied:

$$\log(1 + \text{grossapproval}), \quad \log(1 + \text{thirdpartydollars}).$$

A leverage-related feature is constructed:

$$\text{thirdparty_ratio} = \frac{\text{thirdpartydollars}}{\text{grossapproval} + 1}.$$

This variable is capped to control extreme outliers.

To capture operational efficiency, we define:

$$\text{disbursement_lag_days} = \log(1 + (\text{firstdisbursementdate} - \text{approvaldate})).$$

This serves as a proxy for loan processing speed.

Temporal and loan structure features. From the approval date, we extract approval year, approval month and approval quarter. The loan structure features include:

- `long_term_loan`: indicator for maturity ≥ 20 years.
- `terminmonths_multiple_12`: whether the term is a multiple of 12 months.

These variables capture repayment structure and contractual design.

Geographic and borrower history features. The following features are constructed:

- `diff_lb_state`: lender state $\neq$ borrower state.

- `diff_pb_state`: project state $\neq$ borrower state.
- `repeat_borrower`: whether the borrower has previous loans.
- `l2locid_count`: cumulative number of loans per location identifier.

These features provide proxies for borrower experience and relationship intensity.

Macroeconomic variables. State-level GDP is converted into real terms:

$$\text{GDP_y_st}^{\text{real}} = \frac{\text{GDP_y_st}}{\text{CPI}_m} \times 100.$$

We also define two regime indicators:

- `market_stress_m`: indicator for $\text{VIX} > 25$.
- `yield_inversion_m`: indicator for inverted yield curve, i.e. `yield_spread_m` < 0 . This is widely interpreted as a signal of future economic downturns and increased credit risk.

Categorical encoding. Categorical variables are transformed using one-hot encoding with the first category dropped to avoid multicollinearity. All encoding is performed before dataset splitting to ensure consistent feature spaces across train, validation, and test sets.

Data cleaning. Observations from regions with insufficient sample sizes (PR, VI, GU) are removed to improve statistical stability. Irrelevant or leakage-prone columns (e.g., dates, identifiers, raw monetary variables) are dropped after feature construction.

5.3 Train-Validation-Test Split

Data is split chronologically based on approval year:

- Train: 1990–2011.
- Validation: 2012–2014.
- Test: 2015–2025.

This time-based split prevents forward-looking bias and allows evaluation under realistic distribution shifts.

Feature scaling. Continuous variables are standardized using Z-score normalization:

$$x^{\text{scaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

The mean and standard deviation are computed on the training set and applied to validation and test sets.

Binary variables and target variables are excluded from scaling.

5.4 Model-Specific Target Construction

Binary model. The binary model uses CHGOFF vs PIF observations only, with the target variable defined as default.

Static survival model. For survival modeling, we define:

- `duration`: from first disbursement to event or censoring.
- `event`: 1 for default, 0 for censored (PIF or EXEMPT).

Observations are filtered to ensure:

$$0 < \text{duration} \leq 20 \text{ years}$$

Time-varying survival model. The time-varying dataset expands each loan into monthly observations. Additional variables include:

- month_start, month_stop.
- loan_age.
- event_t (event indicator at each time step).

This structure enables modeling of dynamic hazard rates over time.

Finally, after the feature engineering pipeline, we produce separate datasets for each modeling task:

- Binary: `data_binary_*`
- Survival: `data_survival_*`
- Time-varying survival: `data_tv_*`

Each dataset is fully processed and ready for direct input into downstream models.

6 Binary Response Models

6.1 Overview

The goal of the binary classification task is to predict whether a loan defaults. The target variable is constructed as

$$y = \begin{cases} 1 & \text{if loan status is CHGOFF} \\ 0 & \text{if loan status is PIF} \end{cases}.$$

Observations with status EXEMPT are excluded.

We use the processed static dataset with one observation per loan. The data is split chronologically:

Table 6.1: Time-based data split.

Dataset	Number of Loans	Default Rate
Train (1990–2011)	97,986	11.57%
Validation (2012–2014)	12,898	2.25%
Test (2015–2025)	11,380	1.75%

A combined training dataset (train + validation) is used for model development.

All the models are evaluated using accuracy (ACC), AUC (Area Under ROC Curve), and ROC curve.

6.2 Baseline Models

We first compare four baseline models without any feature selection or hyperparameter tuning.

Table 6.2: Baseline model performance.

Model	Train AUC	Test AUC
Logistic Regression	0.7771	0.5057
LightGBM	0.8390	0.5410
XGBoost	0.8927	0.5390
Random Forest	1.0000	0.5223

Baseline models show limited generalization performance, especially under out-of-time testing. A clear gap between train and test AUC indicates significant overfitting, particularly for tree-based models. This motivates us to conduct further feature selection and parameter tuning work.

6.3 Improved Models with Feature Selection and Hyperparameter Tuning

In this section, we perform grid search using time-series cross-validation with `TimeSeriesSplit(n_splits=5)`. Unlike standard cross-validation, which randomly shuffles data, time-series cross-validation preserves temporal order to avoid look-ahead bias.

Specifically, the data are split into five sequential folds. At each step, the training set consists of all observations up to a certain time point, and the validation set consists of the immediately following block of data. As the split progresses, the training window expands while always preceding the validation window. Formally, if the data are indexed by time $t = 1, \dots, T$, the splits follow the pattern:

$$\text{Train}_k = \{1, \dots, t_k\}, \quad \text{Valid}_k = \{t_k + 1, \dots, t_{k+1}\}, \quad k = 1, \dots, 5.$$

This expanding-window scheme ensures that each model is trained only on past information and evaluated on future data, which better reflects the real-world forecasting setting and helps mitigate overfitting due to temporal leakage.

Feature selection procedure.

1. Time-series CV training. For each split generated by `TimeSeriesSplit`, train a LightGBM model using only past data to avoid look-ahead bias.
2. Feature importance extraction. After each training step, record the feature importance scores (based on split gain or frequency) from the fitted model.
3. Stability via averaging. Aggregate feature importances across all splits by taking the average, which reduces variance and highlights consistently important features rather than split-specific noise.
4. Ranking and selection. Rank features according to their averaged importance and select the top features based on cumulative importance thresholds for downstream modeling (Figure 6.1).

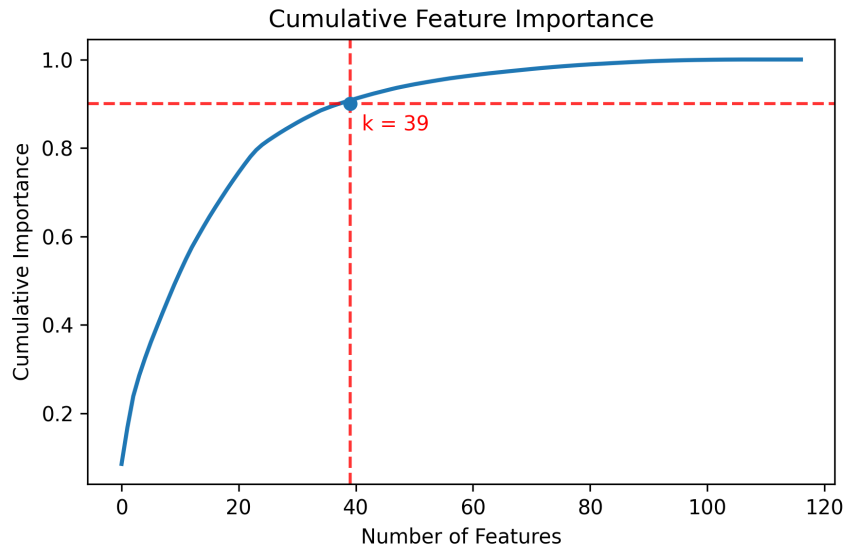


Figure 6.1: Cumulative feature importance.

Figure 6.2 presents the top 15 features ranked by their averaged importance from time-series cross-validation. The most important variables are `disbursement_lag_days`, `log_grossapproval`, and `l2locid_count`.

The prominence of `disbursement_lag_days` suggests that the timing between approval and disbursement contains important information about loan risk, potentially reflecting underwriting complexity or borrower conditions. Similarly, `log_grossapproval` captures loan size, indicating that exposure magnitude is a key driver of default risk.

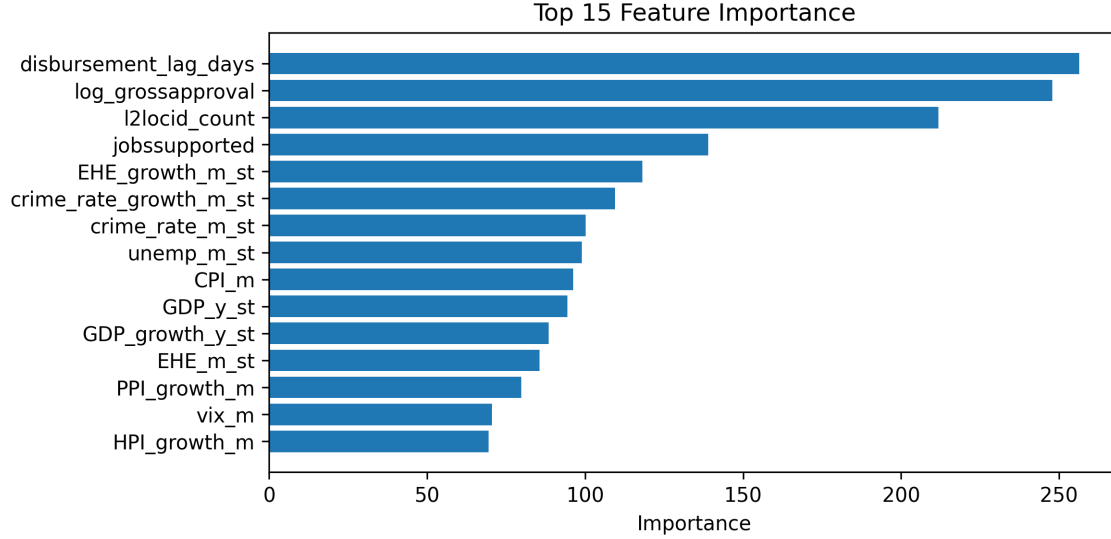


Figure 6.2: Top feature importance ranking.

The variable `l2locid_count`, constructed as the cumulative number of loans within the same geographic location (`l2locid`), reflects local lending intensity and borrower concentration. Its high importance suggests that geographic clustering and local credit conditions play a significant role in predicting defaults.

Overall, the top features highlight that both loan-specific characteristics (size, timing), local market conditions (loan concentration), and borrower scale (employment impact) jointly contribute to predictive performance, with macroeconomic variables providing additional but secondary explanatory power.

Hyperparameter tuning procedure. We perform grid search using time-series cross-validation. The best parameters for each model are shown in Table 6.3.

Table 6.3: Best hyperparameters.

Model	Best Parameters
Logistic Regression	$C=10$, $l1_ratio=1$, $penalty=elasticnet$
LightGBM	$learning_rate=0.05$, $max_depth=5$, $n_estimators=50$, $num_leaves=63$
XGBoost	$learning_rate=0.1$, $max_depth=3$, $n_estimators=100$
Random Forest	$max_depth=10$, $min_samples_split=2$, $n_estimators=300$

After feature selection and tuning, all models exhibit clear improvements in out-of-sample performance. The gains are particularly pronounced for tree-based models, with Random Forest and XGBoost showing substantial increases in test AUC, suggesting that the refined feature set better captures predictive structure while reducing noise. LightGBM also benefits meaningfully, while Logistic Regression shows a more modest improvement, indicating limited capacity to exploit nonlinear relationships.

Table 6.4: Test AUC comparison before and after feature engineering.

Model	Baseline Test AUC	Improved Test AUC
Logistic Regression	0.5057	0.5214
LightGBM	0.5410	0.6107
XGBoost	0.5390	0.6274
Random Forest	0.5223	0.6527

Overall, the results suggest that feature selection enhances generalization and that more flexible models are better able to leverage the improved feature set. Nevertheless, the remaining performance gap indicates that

distribution shift between training and test periods is still a key challenge, which we address in the next section.

6.4 Distribution Shift Adjustment

Covariate shift. From a statistical perspective, *covariate shift* refers to the situation where the distribution of features changes between training and test sets, while the conditional relationship remains stable:

$$P_{\text{train}}(X) \neq P_{\text{test}}(X), \quad \text{but} \quad P(Y | X) \text{ is unchanged.}$$

In our setting, this arises naturally due to the time-based split, where loan characteristics and macro conditions evolve over time.

To diagnose covariate shift, we train a domain classifier to distinguish training vs. test samples using features that are not directly time-indexed. The classifier achieves a very high AUC of 0.9512, indicating that the two distributions are highly separable. This provides strong evidence of substantial covariate shift.

Feature importance from the domain classifier (Figure 6.3) shows that the most important variables are `log_grossapproval`, `disbursement_lag_days`, `jobssupported`, and `log_thirdpartydollars`. These variables capture loan size, disbursement timing, and borrower scale, suggesting that the composition of loans has changed significantly over time, even after adjusting for inflation (i.e., using real values), indicating that the shift is not merely driven by nominal price changes but reflects underlying structural differences in loan characteristics. In contrast, industry and location indicators (NAICS codes and state dummies) are less important, indicating relatively stable structural composition.

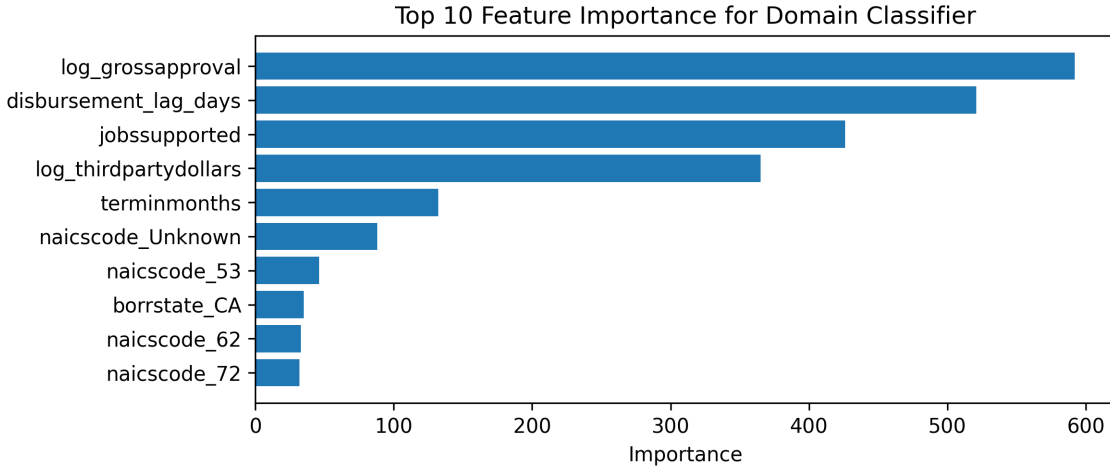


Figure 6.3: Domain classifier feature importance.

We then correct for covariate shift using importance weighting:

$$w_{\text{cov}}(x) = \frac{P_{\text{test}}(x)}{P_{\text{train}}(x)} \approx \frac{\hat{p}_{\text{test}}(x)}{1 - \hat{p}_{\text{test}}(x)},$$

where $\hat{p}_{\text{test}}(x)$ is estimated from the domain classifier. The weights are clipped to stabilize training.

Label shift. In addition to covariate shift, we also observe *label shift*, where the marginal distribution of the outcome changes:

$$P_{\text{train}}(Y) \neq P_{\text{test}}(Y).$$

As shown in Table 6.5, the default rate decreases substantially from 11.57% in the training set to 1.75% in the test set, indicating a strong shift in class balance.

Table 6.5: Label shift statistics.

Metric	Value
Train default rate	0.1157
Test default rate	0.0175
Default weight	0.1512
Non-default weight	1.1111

To correct for this, we apply label reweighting:

$$w_1 = \frac{\pi_{\text{test}}}{\pi_{\text{train}}}, \quad w_0 = \frac{1 - \pi_{\text{test}}}{1 - \pi_{\text{train}}},$$

and assign weights to each observation based on its label.

Final reweighting. Finally, we combine both corrections by multiplying the two weights:

$$w_{\text{final}} = w_{\text{cov}} \times w_{\text{label}},$$

and use these weights to refit the model.

Table 6.6 shows that after applying covariate and label reweighting, all models achieve substantial improvements in out-of-sample performance, with test AUC increasing markedly across the board. The gains are particularly significant for tree-based models, where LightGBM, XGBoost, and Random Forest all exceed 0.70 AUC, indicating that correcting for distribution shift effectively recovers predictive signal that was previously obscured.

Table 6.6: Test AUC comparison: baseline, feature engineering, and distribution shift adjustment.

Model	Baseline Test AUC	Improved Test AUC	Shift-adjusted Test AUC
Logistic Regression	0.5057	0.5214	0.5795
LightGBM	0.5410	0.6107	0.7114
XGBoost	0.5390	0.6274	0.7050
Random Forest	0.5223	0.6527	0.7203

Compared to the improvements from feature engineering alone, reweighting leads to a larger performance boost, highlighting that distribution mismatch is the primary bottleneck instead of model capacity. Logistic Regression also improves meaningfully, though it still lags behind nonlinear models, suggesting limited ability to capture complex patterns even after adjustment.

Overall, these results confirm that both covariate shift and label shift are major drivers of performance degradation, and that explicitly accounting for them is crucial for achieving robust generalization in time-evolving credit risk data.

Figure 6.4 presents the ROC curves for each model before and after distribution shift adjustment, where a clear upward shift can be observed, indicating improved discrimination ability across all classifiers.

Despite these improvements, binary classification models have inherent limitations compared to survival models. In particular, binary models ignore the timing of default events and treat all defaults equally regardless of when they occur, leading to a loss of temporal information. They also cannot naturally handle censoring, which is prevalent in loan data where many observations have not yet defaulted by the end of the sample period. In contrast, survival models explicitly model time-to-event and can incorporate censored observations, providing a more complete and theoretically grounded framework for credit risk analysis, which is detailed in Section 7.

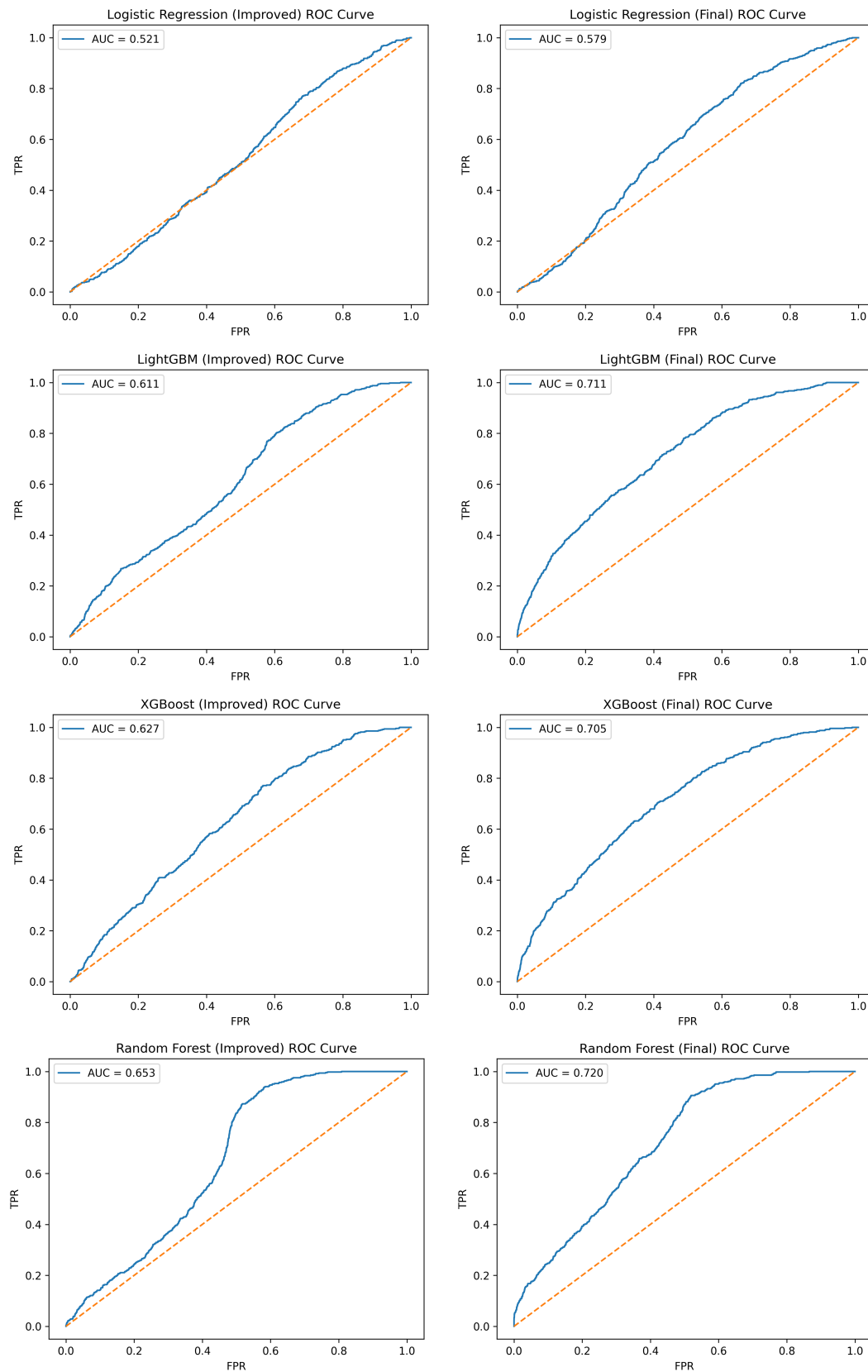


Figure 6.4: ROC curves for improved models (left) and distribution shift-adjusted models (right) across four classifiers. The comparison illustrates the impact of reweighting on predictive performance.

6.5 Interpretable Logistic Regression

We fit a weighted logistic regression using GLM:

$$\log \frac{p}{1-p} = \beta_0 + X\beta.$$

Table 6.7 summarizes selected coefficients, including both loan-level and macro/financial variables.

Table 6.7: Selected logistic regression coefficients.

Variable	Coef	p-value	Interpretation
disbursement_lag_days	0.1030	0.000	Longer delay increases risk
jobssupported	0.1373	0.000	Larger borrower scale increases risk
collateralind_Y	-0.6014	0.000	Collateral reduces risk
CPI_m	-1.5034	0.000	Higher price level reduces risk
GDP_y_st	-0.8960	0.002	Strong economy lowers risk
vix_m	0.1309	0.003	Market volatility increases risk
HPI_growth_m	-0.1079	0.001	Housing growth reduces risk
tnx_m	-0.1728	0.021	Higher rates reduce risk

Among loan-level variables, longer disbursement delays and larger borrower scale (**jobssupported**) are associated with higher default risk, while the presence of collateral significantly reduces risk, consistent with standard credit risk intuition.

Macro and financial variables also play an important role. Higher GDP and housing price growth are associated with lower default risk, reflecting improved economic conditions. In contrast, higher market volatility (VIX) increases risk, capturing financial stress. Interestingly, higher CPI and Treasury yields are associated with lower default risk, which may reflect their correlation with broader economic cycles or monetary conditions.

Overall, the results highlight that both micro-level loan characteristics and macroeconomic conditions jointly determine default risk, reinforcing the importance of incorporating macro variables into the modeling framework.

7 Survival Models

7.1 Overview

The objective of the survival analysis component is to model the timing of loan outcomes rather than only whether an outcome occurs. In the SBA 504 loan setting, two exit events are economically important:

- Default / charge-off.
- Pay in full.

A survival framework is preferable to a static binary classifier for three reasons. First, it preserves the timing information of the event. Second, it naturally handles right censoring, which is substantial in this dataset because many loans are still active at the end of the observation window. Third, it allows us to model dynamic risk through time-varying covariates, especially macroeconomic and market variables that evolve during the life of a loan.

This section develops two layers of models:

1. a static Cox proportional hazards model, used as an interpretable baseline and for feature screening;
2. a time-varying Cox model, which incorporates monthly evolving macro and market covariates and is fit separately for default and pay-in-full events.

The core upgrade in this part of the project is that the time-varying pipeline is fully unified and function-based, and supports both *time to default* and *time to pay in full* under the same preprocessing and fitting framework.

7.2 Kaplan-Meier Exploration

Before imposing any parametric or semi-parametric structure, we first examine nonparametric survival patterns using Kaplan-Meier (KM) curves. This provides a descriptive diagnostic for whether borrower and loan characteristics visibly separate survival trajectories.

Figure 7.1 gives the overall KM curve, and the stratified KM curves by `repeat.borrower`, `ifthirdparty`, `iffranchise`, `borrstate_CA`, and `diff_lb_state`.

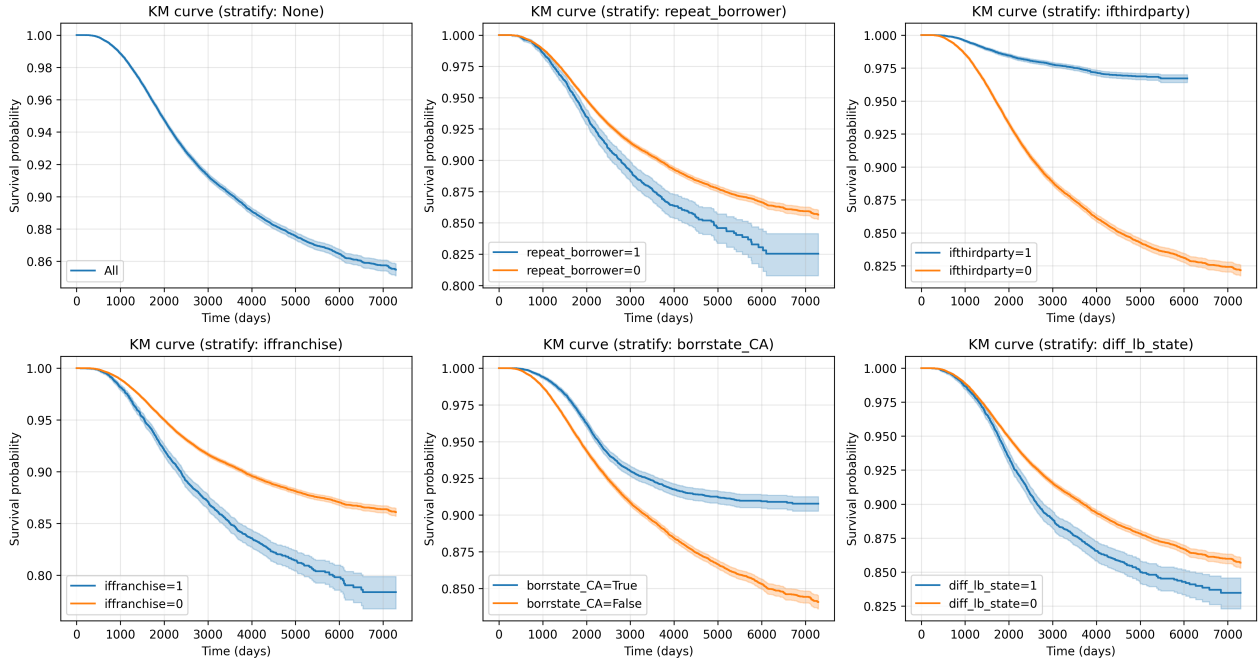


Figure 7.1: Kaplan-Meier curves for the training/validation sample, overall and stratified by selected borrower and loan characteristics.

Clear heterogeneity emerges across different groups. First, repeat borrowers exhibit consistently lower survival probabilities than first-time borrowers, suggesting higher default risk, possibly due to adverse selection or persistent financing needs. Second, loans involving third-party lenders show substantially higher survival rates, indicating that third-party participation plays an important role in screening and monitoring, thereby reducing default risk.

We also observe that franchise loans have lower survival probabilities compared to non-franchise loans, which may reflect their concentration in more cyclical industries such as retail and food services. In terms of geographic effects, loans issued in California exhibit higher survival probabilities than those in other states, suggesting regional differences in economic conditions or borrower quality. Finally, loans where the borrower and lending bank are located in different states show lower survival probabilities, consistent with increased information asymmetry and monitoring costs in cross-state lending.

Overall, these results highlight strong heterogeneity and time-dependent risk patterns across different segments. The persistent separation of survival curves suggests that both loan-level characteristics and structural factors play a crucial role in determining default risk, further motivating the use of survival models that can capture dynamic risk evolution over time.

7.3 Static Cox Models

Loan-level survival dataset. The event variable is coded as:

$$\text{event} = \begin{cases} 1 & \text{default / charge-off} \\ 2 & \text{pay in full} \\ 0 & \text{right censored} \end{cases}.$$

For the static default model, we define

$$\text{event_default} = \mathbf{1}\{\text{event} = 1\},$$

and for the static pay-in-full screening model, we define

$$\text{event_payinfull} = \mathbf{1}\{\text{event} = 2\}.$$

The train/validation and test sample sizes in the loan-level survival dataset are:

Table 7.1: Loan-level survival dataset and event distribution.

Dataset	Total Loans	Defaults	Pay in Full	Censored
Train/Validation	124,294	11,612	98,654	14,028
Test	57,399	199	11,181	46,019

7.3.1 Time to Default Model

We first fit a static Cox proportional hazards model for *time to default*. In this model, only non-time-varying covariates are used. The duration variable is `duration_days`, and the event indicator is `event_default`.

Feature selection. Variable selection is carried out using an L1-penalized Cox model:

$$\text{CoxPHFitter}(\text{penalizer}=0.01, \text{ l1_ratio}=1.0).$$

After fitting, variables with coefficient magnitude greater than 10^{-5} are retained. Then the final static Cox model is re-fit without L1 selection using only the selected variables. Table 7.2 summarizes the model results.

Table 7.2: Static Cox model results for time to default.

Variable	Coef	Hazard Ratio	SE	z	p-value
<code>ifthirdparty</code>	-1.78	0.17	0.04	-48.97	< 0.005
<code>log_grossapproval</code>	0.18	1.20	0.01	18.87	< 0.005
<code>borrstate_CA</code>	-0.40	0.67	0.03	-14.74	< 0.005
<code>borrstate_FL</code>	0.58	1.79	0.03	18.06	< 0.005
<code>naicscode_44</code>	0.32	1.37	0.03	10.55	< 0.005
<code>naicscode_62</code>	-0.40	0.67	0.04	-11.01	< 0.005
<code>naicscode_72</code>	0.76	2.14	0.02	31.36	< 0.005
<code>naicscode_Unknown</code>	-0.86	0.42	0.03	-24.73	< 0.005

Interestingly, the coefficient on `ifthirdparty` is strongly negative, with a hazard ratio of only 0.17, indicating that loans involving third-party lenders experience substantially lower default risk. This effect is both statistically and economically significant, and stands out as the most important driver in the model. This finding is consistent with the right panel of Figure 7.2, where loans with third-party participation exhibit persistently higher survival probabilities over time.

Loan size, proxied by `log_grossapproval`, has a positive coefficient (hazard ratio 1.20), suggesting that larger loans are associated with higher default risk. This may reflect greater exposure or riskier borrower profiles for larger financing amounts.

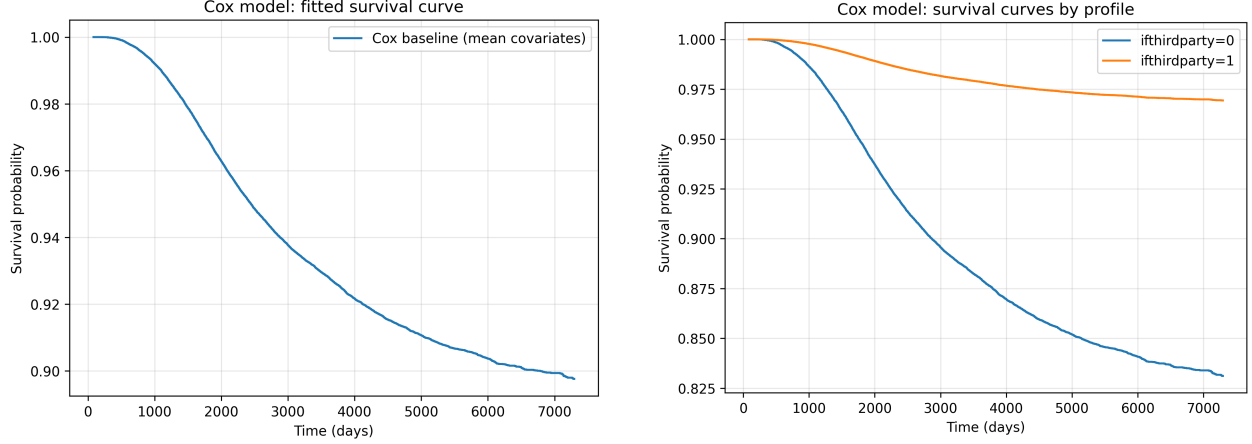


Figure 7.2: Fitted survival curve from the static Cox model for default.

Industry effects are also pronounced. In particular, `naicscode_72` (accommodation and food services) has a hazard ratio of 2.14, indicating significantly higher default risk, consistent with the cyclical and vulnerable nature of this sector.

Finally, geographic heterogeneity is evident. Loans in Florida (`borrstate_FL`) have a hazard ratio of 1.79, implying elevated default risk relative to the baseline, while other regions such as California (not discussed here) show the opposite effect.

Model evaluation. The static Cox model for default is evaluated in two ways:

- horizon-specific ROC/AUC based on predicted survival probability;
- concordance index (C-index).

For a prediction horizon h , we compute the predicted default probability as

$$\hat{p}_h(X) = 1 - \hat{S}(h | X),$$

where $\hat{S}(h | X)$ is the predicted survival probability from the Cox model.

The evaluation mask includes:

- loans that default by horizon h ;
- loans that survive beyond horizon h .

This avoids contaminating the horizon-specific ROC calculation with loans that are censored or exit before the horizon without being valid controls.

The horizon-specific test AUCs are shown in Table 7.3.

Table 7.3: Static Cox default prediction performance across horizons.

Prediction Horizon	Test AUC
1 year	0.7057
3 years	0.5453
5 years	0.5496
7 years	0.5496
10 years	0.6752

The ROC curves across horizons are shown in Figure 7.3.

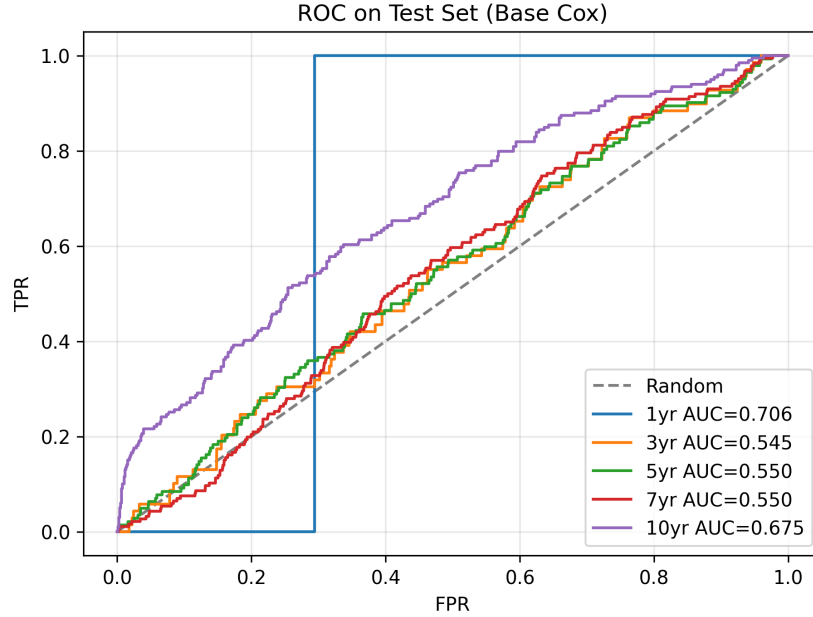


Figure 7.3: Test-set ROC curves for the static Cox default model across multiple horizons.

The concordance statistics are shown in Table 7.4.

Table 7.4: Static Cox concordance indices for time to default.

Metric	Value
Train/Validation C-index	0.7147
Test C-index	0.5684

Overall, the static Cox model provides an interpretable and useful baseline. It has some short-horizon predictive value, but performance weakens at medium and long horizons. This is consistent with the idea that static borrower-level covariates alone are insufficient to capture the evolution of default risk over the life of long-maturity SBA loans.

7.3.2 Time to Pay in Full Model

The notebook also fits a static Cox model for *time to pay in full*, but its purpose is explicitly limited to feature selection, aiming to reduce the time complexity of fitting the later time-varying pay-in-full model. It is not presented as the final economic model for pay-in-full timing.

The same static Cox setup is used, but the event indicator becomes `event_payinfull`. Feature selection is done with:

```
CoxPHFitter(penalizer=0.025, l1_ratio=1.0).
```

The selected variables are:

```
terminmonths, log_grossapproval, thirdparty_ratio, repeat_borrower,
l2locid_count, naicscode_62, naicscode_Unknown, collateralind_Unknown.
```

This screening step serves a practical purpose. The time-varying data is very large after monthly panel expansion, so reducing the number of static borrower-level features before fitting the time-varying pay-in-full model lowers computational cost and improves numerical stability.

7.4 Time-Varying Cox Models

Loan-month-level survival dataset. The time-varying Cox model is fit on monthly panel data with sizes:

Table 7.5: Time-varying survival panel size.

Dataset	Rows	Unique Loans	Exit Events Recorded
TV Train/Validation	14,489,833	125,027	event-specific
TV Test	2,990,679	57,399	event-specific

For event-specific fitting, we convert

$$\text{event_t} = \mathbf{1}\{\text{current row ends in the modeled event}\},$$

so that one time-varying Cox model is fit for default and another for pay in full.

7.4.1 Unified Fitting Pipeline

Candidate features. The monthly macro and market covariates considered are:

GDP_growth_y_st, CPI_growth_m, crime_rate_m_st, crime_rate_growth_m_st,
 EHE_m_st, EHE_growth_m_st, HPI_m, HPI_growth_m, PPI_m, PPI_growth_m, unemp_m_st, unemp_growth_m_st,
 sp500_ret_m, vix_m, tnx_m, t2y_m, russell2000_ret_m, yield_spread_m.

For the default model, these monthly variables are combined with the static default-screened features:

ifthirdparty, log_grossapproval, borrstate_CA, borrstate_FL,
 naicscode_44, naicscode_62, naicscode_72, naicscode_Unknown.

For the pay-in-full model, they are combined with the pay-in-full screened static features:

terminmonths, log_grossapproval, thirdparty_ratio, repeat_borrower,
 l2locid_count, naicscode_62, naicscode_Unknown, collateralind_Unknown.

Thus, each time-varying model begins from 26 candidate covariates.

Cox fitting pipeline. We then implement an event-specific time-varying Cox fitting procedure that can be applied to both models:

1. fit an L1-penalized `CoxTimeVaryingFitter` to select useful variables among the 26 candidate variables;
2. if the L1 fit fails at penalizer 0.005, retry with penalizer 0.01;
3. keep coefficients with absolute value greater than 10^{-5} ;
4. if too few variables survive, use the top absolute coefficients as a fallback;
5. refit the final time-varying Cox model on the selected variables with penalizer 0.01.

Selected features. As shown in Table 7.6, the selected features for the two time-varying Cox models exhibit relatively limited overlap, indicating that the drivers of default and pay-in-full (PIF) outcomes are quite different. While a small number of variables (e.g., `naicscode_62` and `GDP_growth_y_st`) appear in both models, most features are outcome-specific.

The default model is more influenced by loan structure, industry, and certain macro conditions, whereas the PIF model relies more heavily on borrower characteristics and broader financial indicators. This divergence suggests that default and early repayment are governed by distinct underlying mechanisms, supporting the use of competing risk framework.

Table 7.6: Selected variables from time-varying Cox models for default and pay-in-full (PIF).

Variable	Default Model	PIF Model
naicscode_72	✓	
ifthirdparty	✓	
borrstate_FL	✓	
unemp_m_st	✓	
naicscode_Unknown	✓	
t2y_m	✓	
naicscode_62	✓	✓
naicscode_44	✓	
tnx_m	✓	
GDP_growth_y_st	✓	✓
repeat_borrower		✓
collateralind_Unknown		✓
vix_m		✓
russell2000_ret_m		✓
HPI_growth_m		✓
PPI_growth_m		✓
CPI_growth_m		✓
yield_spread_m		✓

7.4.2 Time to Default Model

Table 7.7 shows the model summary results.

Table 7.7: Time-varying Cox model results for time to default.

Variable	Coef	Hazard Ratio	SE	z	p-value
naicscode_72	0.77	2.15	0.02	32.29	< 0.005
ifthirdparty	-2.03	0.13	0.04	-51.86	< 0.005
borrstate_FL	0.57	1.77	0.03	17.75	< 0.005
unemp_m_st	0.20	1.22	0.01	18.95	< 0.005
naicscode_Unknown	0.10	1.10	0.04	2.54	0.01
t2y_m	-0.20	0.82	0.03	-7.90	< 0.005
naicscode_62	-0.51	0.60	0.04	-13.75	< 0.005
naicscode_44	0.35	1.42	0.03	11.74	< 0.005
tnx_m	-0.60	0.55	0.02	-29.33	< 0.005
GDP_growth_y_st	-0.02	0.98	0.01	-2.07	0.04

Compared to the static model, the time-varying Cox results reinforce several key findings while highlighting the role of evolving macroeconomic conditions (Figure 7.4). In particular, **ifthirdparty** remains the most influential variable, with an even stronger effect (hazard ratio 0.13), indicating that third-party participation substantially reduces default risk over time.

Industry effects remain pronounced, with **naicscode_72** exhibiting a hazard ratio above 2, confirming that sectors such as accommodation and food services are significantly more vulnerable to default. Similarly, loans in Florida (**borrstate_FL**) continue to show elevated risk.

Importantly, the time-varying model introduces macroeconomic dynamics. Higher unemployment (**unemp_m_st**) increases default risk, while higher Treasury yields (**tnx_m**) and GDP growth are associated with lower risk, reflecting improving economic conditions. These results demonstrate that default risk is not only driven by static loan characteristics but also evolves with macroeconomic cycles, highlighting the advantage of time-varying survival models in capturing dynamic risk patterns.

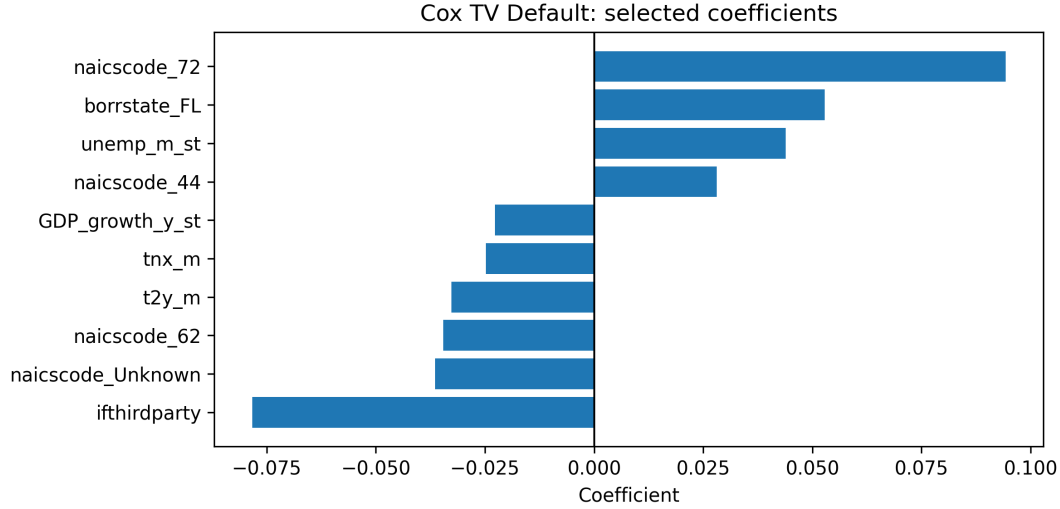


Figure 7.4: Selected coefficients in the time-varying Cox default model.

Figure 7.5 shows that the cumulative hazard rises through time, as expected, and there is a noticeable hazard spike around 240 months. This is consistent with institutional term structure in SBA 504 loans and suggests that maturity-related timing effects remain present even after conditioning on borrower and macro covariates.

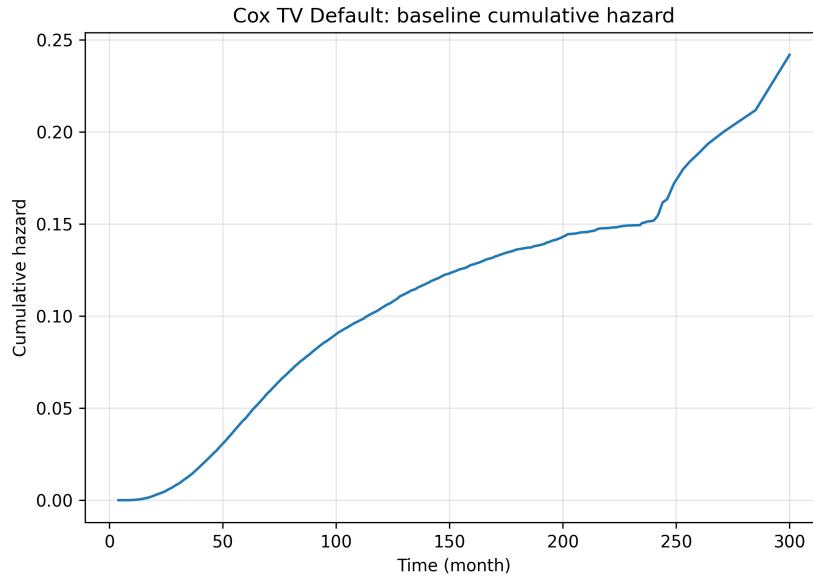


Figure 7.5: Baseline cumulative hazard for the time-varying Cox default model.

Model evaluation. We evaluate the time-varying default model in three ways:

- loan-level horizon-specific ROC/AUC (Table 7.8 & Figure 7.6);
- interval-level C-index and loan-last-interval C-index (Table 7.9).

For the ROC/AUC calculation, we first predicts a partial hazard score for each interval in the test panel. Then we collapse the panel to each loan's last observed interval and computes horizon-specific ROC/AUC using:

`duration_months = month_stop,      event = event_t.`

At a given horizon h months, positive cases are loans that default by h , and controls are loans that remain event-free beyond h .

Table 7.8: Default prediction performance: static Cox versus time-varying Cox.

Horizon	Static Cox AUC	TV Cox AUC	Improvement
1 year	0.7057	0.9302	+0.2245
3 years	0.5453	0.6976	+0.1523
5 years	0.5496	0.7293	+0.1797
7 years	0.5496	0.6924	+0.1428
10 years	0.6752	0.7443	+0.0691

Table 7.9: Concordance comparison for default models.

Metric	Value
Static Cox test C-index	0.5684
TV Cox interval-level C-index	0.6511
TV Cox loan-last-interval C-index	0.7048

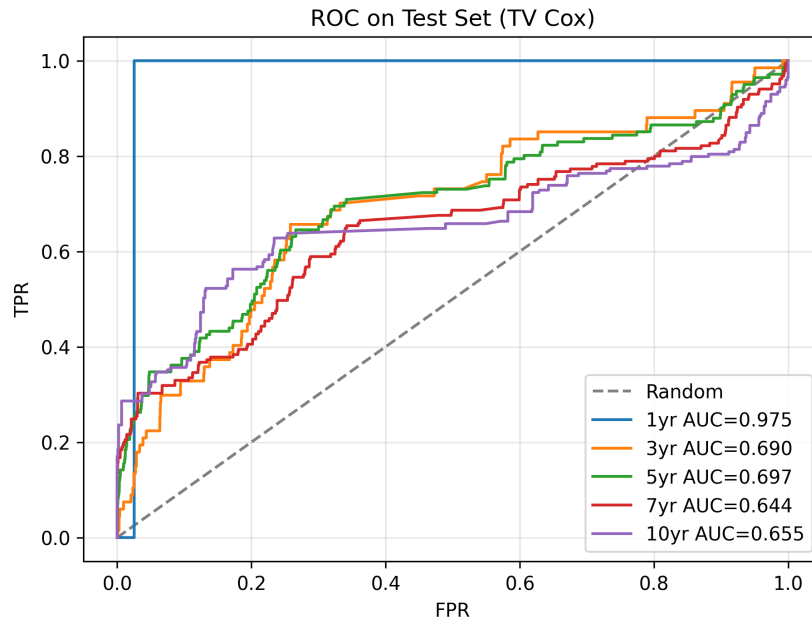


Figure 7.6: Test-set ROC curves for the time-varying Cox default model across multiple horizons.

These results show a clear improvement from incorporating time-varying macro conditions. The gains are especially large at short and medium horizons, which is exactly where evolving macro conditions should matter most for hazard timing. This supports the view that the time-varying Cox model is not merely a robustness check, but a substantially better representation of the default process in SBA 504 loans.

7.4.3 Time to Pay in Full Model

Table 7.10 shows the model summary results.

Table 7.10: Time-varying Cox model results for time to pay-in-full.

Variable	Coef	Hazard Ratio	SE	z	p-value
repeat.borrower	-1.76	0.17	0.02	-108.19	< 0.005
collateralind.Unknown	0.63	1.88	0.01	71.21	< 0.005
vix_m	-0.38	0.68	0.00	-81.97	< 0.005
russell2000_ret_m	-0.34	0.71	0.00	-84.87	< 0.005
HPI_growth_m	0.08	1.09	0.00	21.87	< 0.005
PPI_growth_m	0.16	1.18	0.00	35.58	< 0.005
naicscode_62	-0.09	0.91	0.01	-8.80	< 0.005
CPI_growth_m	-0.03	0.97	0.00	-5.40	< 0.005
GDP_growth_y_st	0.01	1.01	0.00	2.95	< 0.005
yield_spread_m	-0.09	0.91	0.00	-25.16	< 0.005

Compared to the default model, the time-varying Cox results for pay-in-full (PIF) reveal a distinct set of drivers (Figure 7.7). Most notably, **repeat.borrower** emerges as the dominant factor, with a large negative coefficient and a hazard ratio of 0.17. This indicates that repeat borrowers are significantly less likely to repay early, suggesting persistent financing needs or rollover behavior. The magnitude and statistical significance of this effect highlight **repeat.borrower** as the single most important determinant in the PIF model.

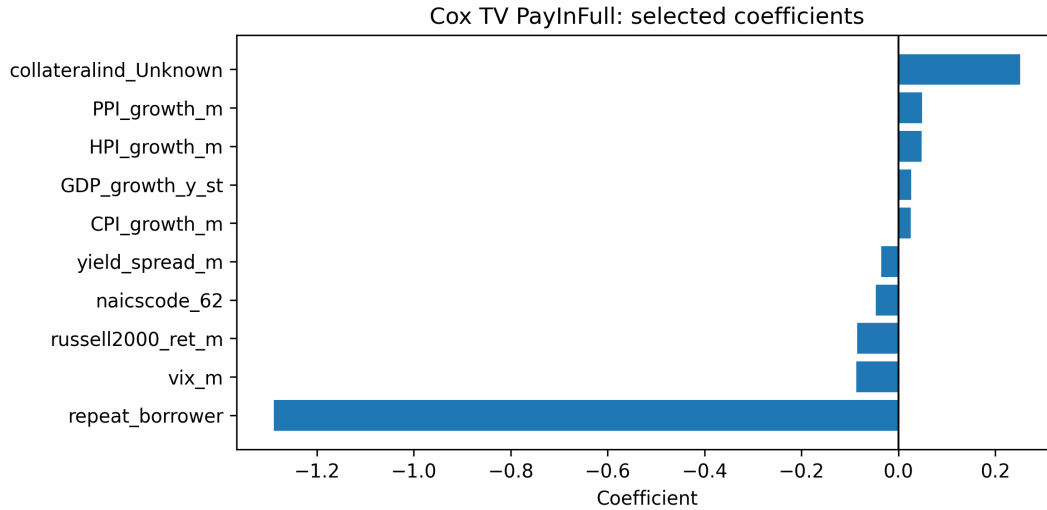


Figure 7.7: Selected coefficients in the time-varying Cox PIF model.

In contrast to the default model, macro-financial variables play a more prominent role. Higher market volatility (**vix_m**) and lower equity returns (**russell2000_ret_m**) are associated with reduced prepayment rates, reflecting tighter financial conditions. Similarly, a flatter yield curve (**yield_spread_m**) reduces the likelihood of early repayment, consistent with weaker economic expectations.

Finally, variables such as **collateralind.Unknown** and price-related indicators (e.g., **PPI_growth_m**) also influence prepayment behavior, suggesting that both borrower characteristics and macroeconomic conditions jointly shape repayment dynamics.

It is worth noting that this model is not intended solely for predicting prepayment behavior in isolation. Instead, it is developed as a complementary component to the default model within a competing risks framework, which is used in the subsequent portfolio loss simulation. By jointly modeling default and pay-in-full events, we can more accurately capture the timing and probability of different exit states, which is essential for realistic exposure and loss dynamics in the simulation.

8 Portfolio Loss Simulation and Tranche Analysis

8.1 Overview

The final stage of the project translates the fitted loan-level survival models into portfolio-level credit loss distributions. The goal is to simulate the losses of a hypothetical portfolio of 1,000 SBA 504 loans randomly sampled from the test set over two investment horizons, 5 years and 10 years, and then evaluate both whole-portfolio risk and tranche-level risk.

Relative to a simpler default-only simulation (what I did in the presentation), this implementation introduces two important upgrades. First, it uses a *competing-risk* time-varying Cox framework, so that default and pay-in-full are modeled as separate but competing exit events. Second, it uses *dynamic EAD*, so loan exposures decline over time through amortization rather than remaining fixed throughout the simulation horizon.

Two pipelines. We evaluate two parallel exposure specifications:

1. Equal-EAD pipeline: each loan is assigned unit exposure, $EAD = 1$, with $LGD = 1$.
2. Dollar-EAD pipeline: each loan is assigned $EAD = \text{grossapproval}$, with $LGD = 0.6$.

This design allows the simulation to distinguish between two different views of risk. The equal-EAD setup emphasizes the *frequency* of default events across loans, while the dollar-EAD setup emphasizes the *economic concentration* of losses by actual loan size.

Why a competing-risk simulation is needed. A one-shot classifier for “default by horizon T ” is not sufficient for portfolio loss simulation. Losses depend not only on whether a loan defaults, but also on *when* it defaults. Early default and late default produce different remaining balances and therefore different loss severities under amortization. In addition, SBA loans may exit the portfolio through PIF before ever defaulting. If prepayment is ignored, the model will overstate the number of loans that remain exposed to future default risk.

Accordingly, the simulation treats each surviving loan in each month as facing three possible outcomes:

Default, Prepay / Pay in Full, Survive to Next Month.

Default and prepayment are absorbing exits. This setup is more realistic than a single-event model because prepayment removes loans from the future default-risk pool and therefore directly changes the portfolio loss distribution.

8.2 Simulation Pipeline

Survival models. The simulation does not generate loan events from ad hoc probabilities. Instead, it reuses the fitted time-varying survival models from the previous section. Two separate cause-specific time-varying Cox models are estimated using the variables selected in Sections 7.4.2 and 7.4.3.

After fitting, we extract each model’s baseline cumulative hazard and stores it as a pair of time and value arrays. These arrays are then converted into interpolation functions, which enables month-ahead transition probabilities to be computed during simulation.

Portfolio construction. The portfolio is built from the test panel. We first sort the panel by `loan_id` and `month_stop`, then take the latest available observation for each loan. This creates a cross-section representing each loan’s most recent state at the evaluation date. The latest observed month in the test panel is also identified, and for a predefined set of macro variables, we compute the median value at that latest month and overwrites the portfolio snapshot with these frozen macro conditions. Hence the simulation evaluates the portfolio “as of today”: loan identities and static borrower attributes are fixed, while current macro conditions are imposed uniformly across the snapshot.

The portfolio size is set to

$$N = 1000.$$

Because the available number of latest-loan observations exceeds 1,000, the portfolio is sampled without replacement in practice.

Exposure specification and dynamic EAD. A key improvement over a fixed-exposure simulation is that each loan’s exposure declines over time through amortization. We define

$$\text{MONTHLY_RATE} = \frac{0.0375}{12},$$

where 0.0375 comes from the latest U.S. interest rate.

If an individual loan’s contractual term is missing, we use the fallback

$$\text{LOAN_TERM_MONTHS} = 240.$$

For loan i , if the original exposure is $B_{i,0}$, elapsed simulation time is t , contractual maturity is T_i , and monthly rate is r , then the remaining-balance factor is

$$\frac{B_i(t)}{B_i(0)} = \frac{(1+r)^{T_i} - (1+r)^t}{(1+r)^{T_i} - 1}.$$

Therefore,

$$B_i(t) = B_{i,0} \cdot \frac{(1+r)^{T_i} - (1+r)^t}{(1+r)^{T_i} - 1}.$$

This factor is clipped to $[0, 1]$. The economic interpretation is straightforward: long-horizon default loss should depend on the *remaining balance at default*, not the origination balance. This matters particularly for the 10-year horizon, because keeping EAD fixed would systematically overstate late-horizon losses.

Month-ahead competing-risk probabilities. At each month, the simulation computes separate cause-specific probabilities for default and prepayment using the fitted Cox models. Let a denote the current loan age in months, and let $\Delta H_D(a)$ and $\Delta H_P(a)$ denote the month-ahead increments in the interpolated baseline cumulative hazards for default and prepayment, respectively. For loan i , with covariate vectors $x_{i,D}$ and $x_{i,P}$, we compute

$$\text{hr}_{i,D} = \exp(x_{i,D}^\top \beta_D), \quad \text{hr}_{i,P} = \exp(x_{i,P}^\top \beta_P).$$

These are then converted into month-ahead probabilities:

$$p_{i,D} = 1 - \exp\{-\Delta H_D(a) \text{hr}_{i,D}\}, \quad p_{i,P} = 1 - \exp\{-\Delta H_P(a) \text{hr}_{i,P}\}.$$

If

$$p_{i,D} + p_{i,P} > 1,$$

we rescale both probabilities proportionally so that the total does not exceed one. Survival probability is then set to

$$p_{i,S} = 1 - p_{i,D} - p_{i,P}.$$

This procedure provides the discrete-time monthly transition probabilities needed for Monte Carlo path simulation.

Monte Carlo algorithm. For each horizon $H \in \{5, 10\}$ years, the number of monthly steps is

$$n_{\text{steps}} = 12H.$$

For each of the

$$\text{NUM_SIMS} = 10,000$$

Monte Carlo trials, we perform the following loop:

1. Start with the portfolio snapshot and store each loan’s original EAD as `EAD_original`.
2. Initialize `months_elapsed=0` for all loans.

3. For each month:

- (a) update the current EAD using the amortization formula;
- (b) compute p_D , p_P , and p_S from the two cause-specific TV-Cox models;
- (c) draw a uniform random variable for each surviving loan;
- (d) classify the loan as default, prepay, or survive;
- (e) add default loss

$$\text{loss} = \text{EAD}(t) \times \text{LGD}$$

for loans that default in that month;

- (f) remove prepaid loans with zero loss;
- (g) carry surviving loans into the next month by incrementing `month_stop` and `months_elapsed`.

The simulation stores total portfolio loss for every scenario. Losses are recorded with a negative sign internally, so larger losses correspond to more negative numbers. The final reported loss rates convert them back to positive values.

An important modeling choice is that macro conditions are frozen at the latest observed month rather than simulated forward through a stochastic macro model. Thus the simulation is a *current-environment conditional risk analysis*, not a macro-scenario generator. This choice matches the project requirement to evaluate a portfolio as of the current date using current macro conditions.

8.3 Risk measures and Tranche Mapping

Risk measures. After simulating scenario losses, we compute portfolio-level and tranche-level risk measures. Let $L^{(m)}$ denote the simulated loss in scenario m . Value-at-Risk and Average Value-at-Risk are computed directly from the empirical distribution.

For confidence level α ,

$$\text{VaR}_\alpha = -Q_{1-\alpha}(L),$$

where $Q_{1-\alpha}(L)$ is the empirical lower-tail quantile of the signed loss sample.

Average Value-at-Risk (expected shortfall) is computed as the average loss in the tail beyond the corresponding quantile:

$$\text{AVaR}_\alpha = -\mathbb{E}[L \mid L \leq Q_{1-\alpha}(L)].$$

The confidence interval for VaR is constructed from binomial order-statistic logic, rather than from a normal approximation.

Tranche mapping. Next, we map portfolio loss rates into tranche loss rates. Two tranches are considered:

$$\text{Mezzanine} : [5\%, 15\%], \quad \text{Senior} : [15\%, 100\%].$$

If the realized portfolio loss rate is ℓ , and a tranche has attachment point a and detachment point d , we compute tranche loss rate as

$$\ell_{\text{tranche}} = \frac{\min\{\max(\ell - a, 0), d - a\}}{d - a}.$$

Thus, losses below the attachment point do not affect the tranche, losses between attachment and detachment are absorbed proportionally, and losses above detachment wipe the tranche out completely.

8.4 Numerical Results

In this section, we present a concise set of key results to illustrate the main findings from the simulation. We focus on a few representative plots and tables that highlight the core patterns in portfolio loss dynamics and tranche performance.

8.4.1 Equal-EAD Pipeline

Table 8.1 reports portfolio-level risk under the equal-EAD specification, where every loan receives unit exposure. Because all loans are equally weighted, this setup reflects how frequently defaults occur in the portfolio rather than how concentrated they are in large exposures.

Portfolio losses increase materially from 5 years to 10 years, as expected.

Table 8.1: Portfolio loss distribution under equal EAD.

Horizon	Mean	VaR95	VaR99	AVaR95	AVaR99
5Y	0.0520	0.0640	0.0690	0.0667	0.0712
10Y	0.0830	0.0980	0.1040	0.1016	0.1070

Table 8.2: Tranche loss distribution under equal EAD.

Tranche	Horizon	Mean	P(any loss)	VaR95
Mezz [5,15]%	5Y	0.0390	0.5803	0.1400
Senior [15,100]%	5Y	0.0000	0.0000	0.0000
Mezz [5,15]%	10Y	0.3298	1.0000	0.4800
Senior [15,100]%	10Y	0.0000	0.0000	0.0000

Table 8.2 shows the corresponding tranche losses. The mezzanine tranche is highly exposed to tail losses, especially at the 10-year horizon. In contrast, the senior tranche remains fully protected in the simulated scenarios. This implies that under a loan-count perspective, the portfolio generates enough defaults to threaten the mezzanine layer, but not enough aggregate loss to reach the senior attachment point.

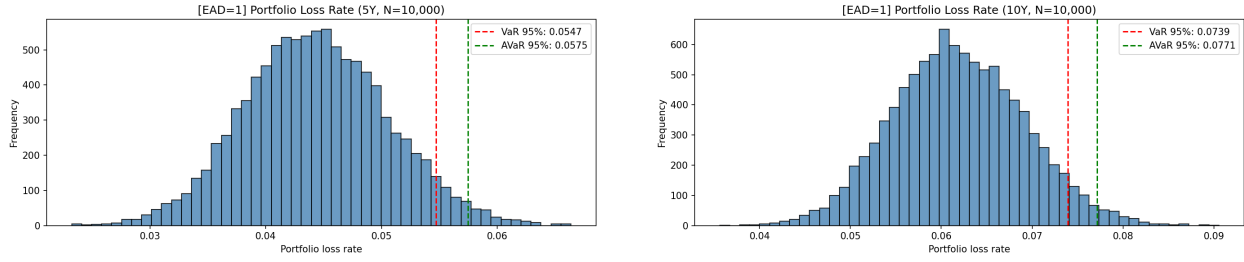


Figure 8.1: Portfolio loss histograms under equal EAD.

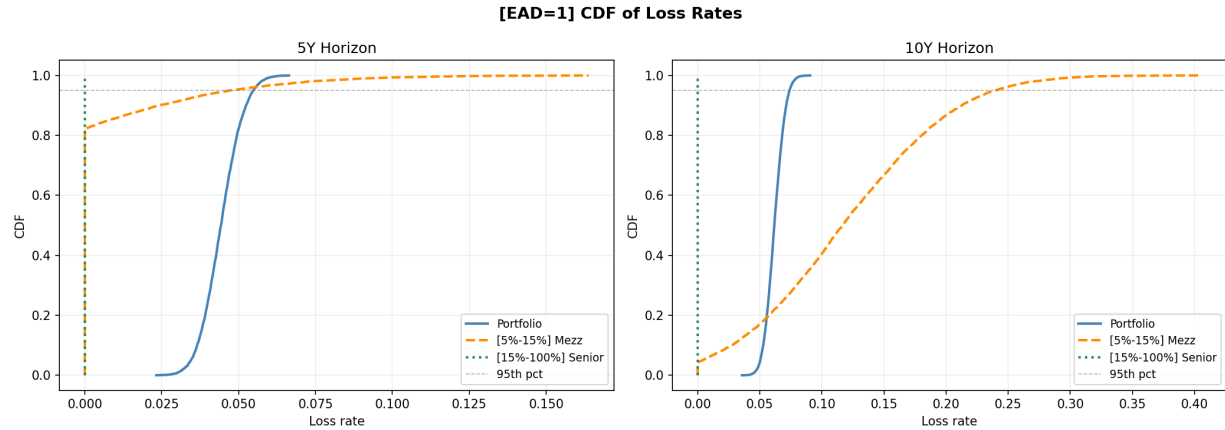


Figure 8.2: CDF comparison of portfolio and tranche loss rates under equal EAD. The left panel shows the 5-year horizon, where losses are highly concentrated and the senior tranche remains largely unaffected. The right panel shows the 10-year horizon, where the loss distribution becomes more dispersed, leading to a substantial increase in mezzanine tranche risk while the senior tranche remains protected except under extreme scenarios.

8.4.2 Dollar-EAD Pipeline

Table 8.3 reports portfolio-level risk under the dollar-EAD specification, where each loan's exposure equals **grossapproval** and default losses are scaled by $LGD = 0.6$. These losses are substantially lower than those in the equal-EAD pipeline. The reason is that defaults are not concentrated proportionally in the largest loans. Once actual loan sizes are used, the portfolio appears less risky in dollar terms than in loan-count terms.

Table 8.3: Portfolio loss distribution under dollar EAD.

Horizon	Mean	VaR95	VaR99	AVaR95	AVaR99
5Y	0.0322	0.0427	0.0478	0.0457	0.0500
10Y	0.0514	0.0645	0.0705	0.0681	0.0735

Table 8.4: Tranche loss distribution under dollar EAD.

Tranche	Horizon	Mean	P(any loss)	VaR95
Mezz [5,15]%	5Y	0.0001	0.0037	0.0000
Senior [15,100]%	5Y	0.0000	0.0000	0.0000
Mezz [5,15]%	10Y	0.0378	0.5601	0.1453
Senior [15,100]%	10Y	0.0000	0.0000	0.0000

Table 8.4 gives the tranche-level results. Under real-dollar exposure, the mezzanine tranche becomes almost riskless at 5 years and remains far safer than in the equal-EAD simulation even at 10 years. The senior tranche again remains essentially untouched.

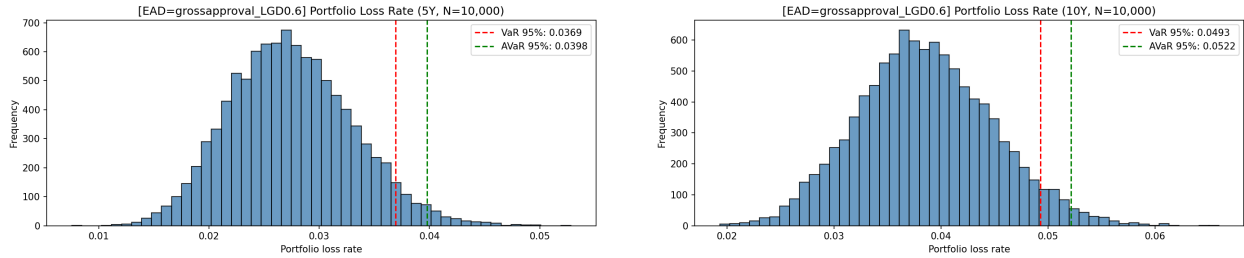


Figure 8.3: Portfolio loss histograms under dollar EAD.

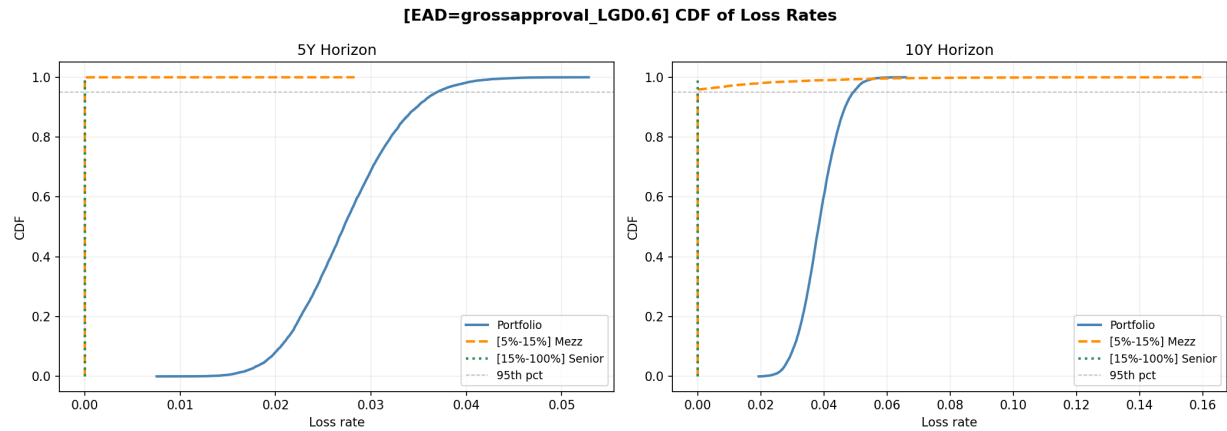


Figure 8.4: CDF comparison of portfolio and tranche loss rates under dollar EAD. The left panel (5-year horizon) shows that losses remain relatively low and concentrated, with almost no impact on mezzanine and senior tranches. In contrast, the right panel (10-year horizon) exhibits a wider loss distribution, where portfolio losses increase and begin to affect the mezzanine tranche, while the senior tranche remains largely insulated except in extreme tail scenarios. Compared to the equal EAD case, incorporating heterogeneous exposures leads to a more concentrated and realistic loss distribution.

8.4.3 Direct Comparison of the Two EAD Specifications

Table 8.5 highlights the most important differences between the two portfolio constructions.

Table 8.5: Comparison of equal EAD and dollar EAD results.

Metric	Equal EAD	Dollar EAD	Interpretation
Mean portfolio loss (5Y)	0.0520	0.0322	Dollar-weighted losses are lower
Mean portfolio loss (10Y)	0.0830	0.0514	Exposure concentration matters
Mean mezz loss (5Y)	0.0390	0.0001	Mezz is highly sensitive to EAD definition
Mean mezz loss (10Y)	0.3298	0.0378	Equal EAD emphasizes default counts

This comparison is one of the most important findings of this section. The risk assessment changes substantially depending on whether loans are weighted equally or by real exposure. Equal EAD answers the question, “How frequently do loans default?” Dollar EAD answers the more economically relevant question, “How much money is actually at risk when defaults occur?” In this portfolio, many simulated defaults occur among smaller loans, so the equal-EAD setup amplifies apparent tranche risk relative to the dollar-EAD setup.

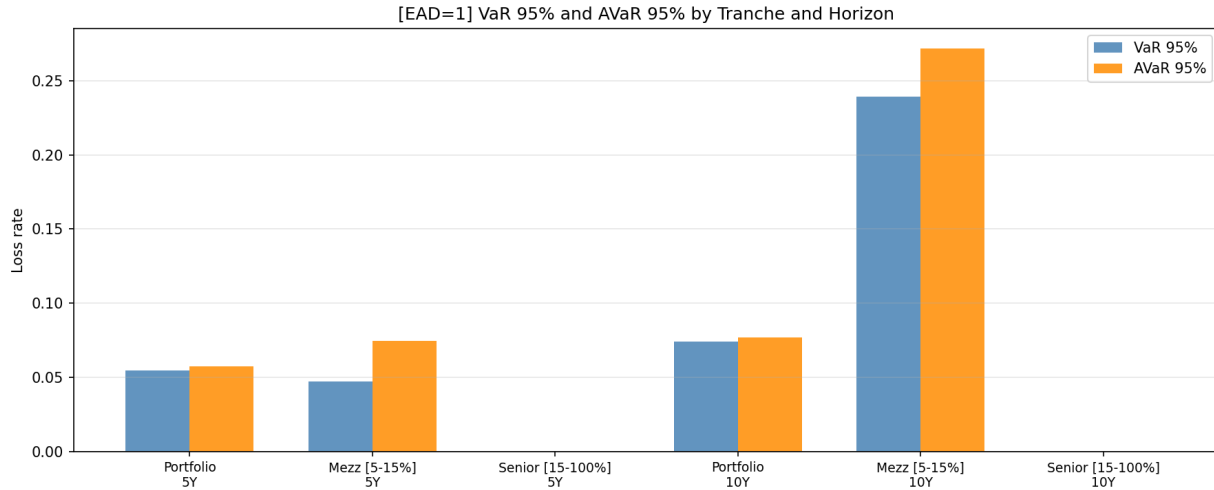


Figure 8.5: Comparison of $\text{VaR}_{95\%}$ and $\text{AVaR}_{95\%}$ across horizons and tranches under equal EAD.

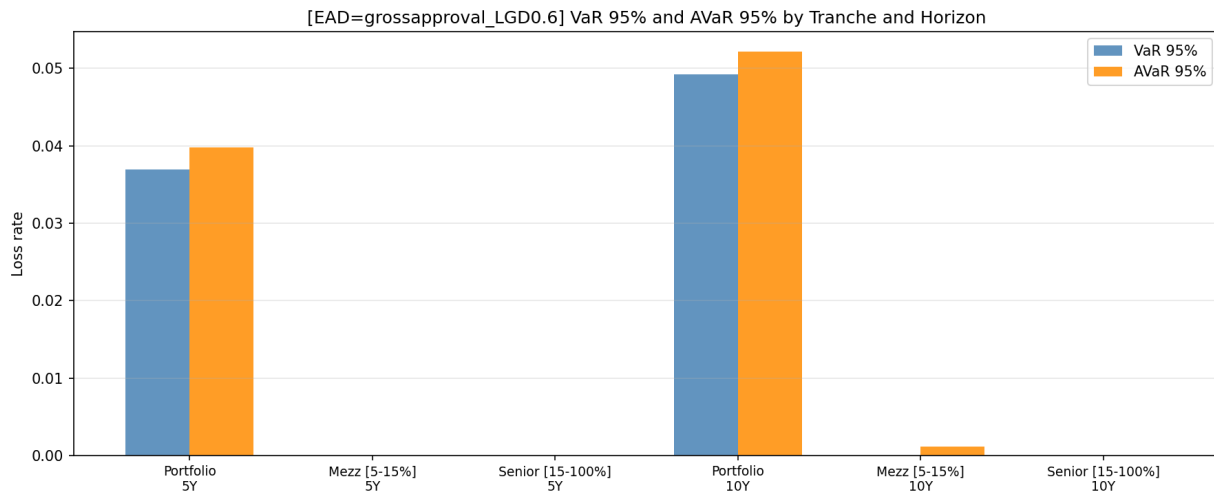


Figure 8.6: Comparison of $\text{VaR}_{95\%}$ and $\text{AVaR}_{95\%}$ across horizons and tranches under dollar EAD.

From a structured-credit perspective, the mezzanine tranche [5%, 15%] acts as the main risk-absorbing layer. It begins taking losses once portfolio loss exceeds 5%, and it is therefore much more sensitive to moderate and tail portfolio losses. In contrast, the senior tranche [15%, 100%] is only affected once portfolio loss exceeds 15%, so it serves as a capital-preserving layer in the simulated scenarios.

The simulation implies three main risk-management conclusions. First, modeling prepayment explicitly matters because it reduces the set of loans that remain exposed to future default. Second, modeling dynamic EAD matters because remaining balances decline over time, especially over a 10-year horizon. Third, tranche conclusions are highly sensitive to the EAD specification. If one uses equal EAD, mezzanine risk appears much larger because defaults are counted rather than weighted by size. If one uses real EAD, the same default pattern translates into much smaller economic loss because large balances are not defaulting in proportion to their share of exposure.

9 Conclusion

This project develops a comprehensive and end-to-end framework for modeling and quantifying credit risk in SBA 504 loans, integrating data engineering, predictive modeling, and portfolio-level risk simulation. Starting from raw loan-level data, the analysis constructs a clean dataset, incorporates macroeconomic variables, and designs a unified feature engineering pipeline that supports multiple modeling approaches. Three complementary model families are implemented: binary classification models to predict default probability, static survival models to capture time-to-event dynamics, and time-varying Cox models to incorporate panel structure and evolving covariates.

A key contribution of the report is the transition from loan-level prediction to portfolio-level risk measurement. Rather than focusing solely on predictive accuracy, the analysis builds a competing-risk simulation framework that jointly models default and prepayment, allowing for realistic evolution of the loan pool over time. By incorporating dynamic exposure-at-default (EAD) through amortization and using Monte Carlo simulation, the framework produces full loss distributions across different horizons. These distributions are then mapped into tranche-level losses, providing a structured credit perspective on portfolio risk.

Overall, the project demonstrates how combining survival analysis with simulation techniques provides a coherent bridge from micro-level credit modeling to macro-level risk management. The framework is flexible and extensible, and can be further enhanced by incorporating stochastic macroeconomic scenarios, loan-level LGD modeling, and more detailed cash-flow waterfall structures.

Contribution

Xiaowei Zeng completed all aspects of the project, including data cleaning, exploratory data analysis, feature engineering, model development, model evaluation, simulation, and presentation.

Code Availability

All codes used in this project are publicly available in the following GitHub repository:

<https://github.com/xw-zeng/Financial-Risk-Analytics-2026Winter>

All experiments were conducted using **Python**, with key libraries including **pandas**, **numpy**, **matplotlib**, **scikit-learn**, and **lifelines**. Random seeds are fixed where applicable to ensure reproducibility of simulation results.